

Multivariate Gait Analysis:

Machine Learning and Deep Learning Approaches to Classifying Joint-Angle Dynamics under Bracing Conditions

Abstract

Accurate classification of human gait patterns under various bracing conditions is crucial for clinical diagnostics, personalized rehabilitation, and assistive technology development. Traditional approaches often lack robustness and struggle to generalize due to high inter-subject variability and the complex temporal-spatial dynamics inherent in gait mechanics. This study addresses these challenges by systematically evaluating and comparing diverse computational approaches—including Random Forest, Hidden Markov Models (HMM), Long Short-Term Memory (LSTM) networks with attention mechanisms, and Spatio-Temporal Graph Convolutional Networks (ST-GCN)—using multivariate joint-angle data captured across unbraced, knee-braced, and ankle-braced conditions. Rigorous feature engineering, exploratory analyses using PCA and UMAP, and sophisticated interpretability methods such as gradient-based saliency and attention-weighted analyses were employed to elucidate biomechanical relevance. Results demonstrate near-perfect accuracy with Random Forest and HMM models, robust temporal pattern recognition with attention-enhanced LSTM networks, and valuable insights into joint compensatory mechanisms through ST-GCN despite overfitting challenges. This comprehensive comparative analysis not only underscores the effectiveness of advanced machine learning and deep learning methods in gait classification but also enhances interpretability, paving the way for patient-centered clinical diagnostics and personalized interventions in healthcare and biomechanics.

1. Introduction

1.1 Problem Statement

Efficient and precise classification of gait patterns can significantly enhance medical diagnostics, rehabilitation planning, and treatment strategies, particularly in conditions affecting mobility such as neurological disorders, orthopaedic injuries, and post-operative recovery [1][2]. However, current methodologies often struggle with robustness and generalization due to variability in gait mechanics between individuals, sensitivity to experimental settings, and noisy or incomplete data [3][4]. Traditional gait analysis approaches primarily rely on simple statistical features or linear modeling techniques, which fail to capture complex temporal dependencies and subtle biomechanical variations that characterize human locomotion [5][6]. Thus, there is a critical need for advanced, robust analytical models capable of capturing and interpreting intricate multivariate and temporal relationships inherent in gait dynamics, ultimately leading to more accurate, individualized assessments and interventions [7] [8].

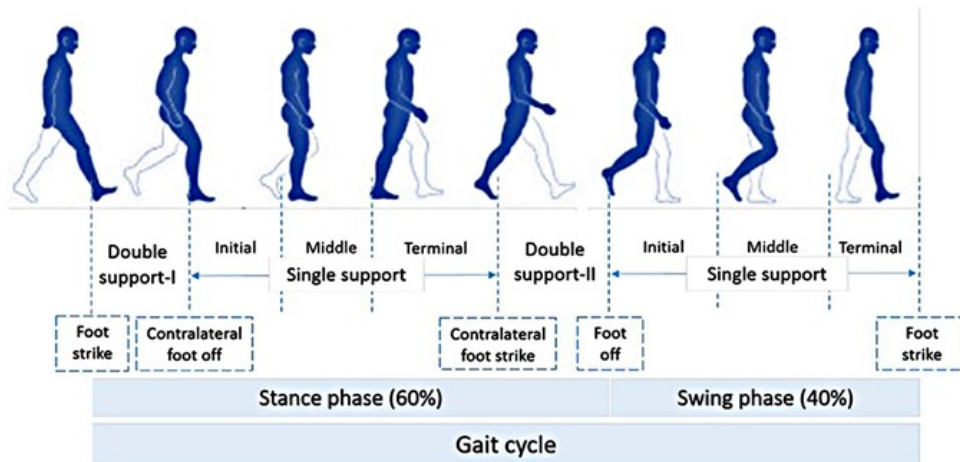


Fig. 1: Normal Gait Cycle [9]

1.2 Motivation and Impact

Accurate gait analysis holds substantial potential to transform healthcare, particularly through early diagnosis and personalized rehabilitation strategies for musculoskeletal and neurological conditions. Early identification of gait abnormalities enables timely clinical interventions, reducing long-term morbidity and improving patient outcomes [10]. Precise characterization

of gait patterns facilitates tailored rehabilitation programs, optimizing recovery trajectories and significantly enhancing quality of life [1].

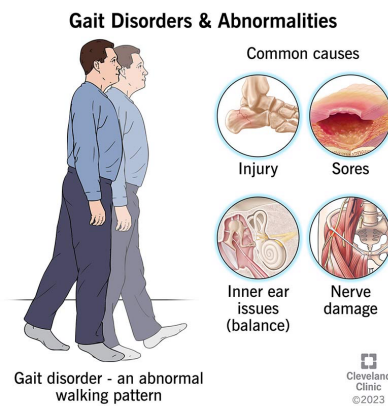


Fig. 2: Gait disorder and common causes [11]

In musculoskeletal healthcare, precise gait analysis aids in diagnosing conditions such as osteoarthritis and sports-related injuries by detecting subtle deviations before they evolve into pronounced impairments [12]. In neurological contexts, gait analysis is instrumental in identifying and monitoring diseases such as Parkinson's, stroke, cerebral palsy, and multiple sclerosis, thus allowing clinicians to monitor disease progression and adjust treatment plans dynamically [13,14].

Moreover, comprehensive gait characterization supports the development of assistive technologies, including advanced orthotic and prosthetic devices, enhancing their personalization and functional efficacy [15]. Consequently, robust analytical models and accurate computational methods in gait pattern analysis not only promise enhanced diagnostic accuracy but also promote individualized, patient-centered healthcare.

1.3 Objectives

This research aims to develop robust gait classification models leveraging diverse computational methodologies, encompassing classical machine learning, temporal modeling techniques, and advanced deep learning approaches. A central objective is the systematic evaluation and comparative analysis of these multiple computational frameworks to determine their effectiveness, accuracy, and robustness in gait classification. Additionally, the research prioritizes enhancing the interpretability and clinical relevance of the models, thereby

facilitating clearer insights into biomechanical phenomena and improving their applicability in areas such as clinical diagnostics and personalized medicine.

1.4 Key Contributions

The study contributes to the field by conducting a detailed comparative analysis of various modelling techniques, including Random Forest classifiers, Hidden Markov Models, Long Short-Term Memory networks, and Spatio-Temporal Graph Convolutional Networks. Another significant contribution is the in-depth interpretability assessment, achieved through sophisticated techniques such as attention-weighted analysis and gradient-based saliency maps. These methods illuminate the biomechanical relevance of model predictions and ensure transparency in clinical decision-making. Furthermore, the research demonstrates rigorous validation practices, employing cross-validation and subject-level testing across diverse conditions and participant cohorts, underscoring the robustness and generalizability of the developed gait classification models.

2. Literature Review

Traditional Machine Learning Approaches

Traditional machine learning methods for gait analysis, such as Random Forest, Support Vector Machines (SVM), and Logistic Regression, have demonstrated considerable utility due to their well-established frameworks, extensive documentation, and proven efficacy in handling structured and tabular datasets [16][17]. These approaches offer straightforward implementation procedures and provide interpretable outcomes, making them suitable for clinical applications where explainability is crucial [18]. However, traditional machine learning methods possess notable limitations, particularly their constrained ability to model complex temporal and spatial dependencies intrinsic to gait dynamics [19]. Consequently, their performance is significantly influenced by the quality of feature engineering and feature selection, requiring extensive domain expertise and pre-processing to achieve robust and generalizable outcomes [20]. Such reliance on manually crafted features poses potential challenges in accurately generalizing these models to diverse gait patterns without thorough preprocessing and tailored adjustments [21].

Deep Learning Approaches

The advent of deep learning (DL) has significantly advanced gait analysis by enabling models to automatically discover complex, hierarchical features directly from raw data, thus reducing reliance on manual feature engineering. Among these methods, Convolutional Neural Networks (CNNs) have demonstrated exceptional proficiency in extracting spatial features from gait patterns due to their powerful capability to capture local structural dependencies in image-like or sensor-based data. In contrast, Long Short-Term Memory (LSTM) networks have gained prominence owing to their inherent strengths in modeling temporal dependencies within sequential gait data, capturing critical dynamics across successive time points effectively [22][23].

Hybrid architectures combining CNNs and LSTMs have emerged as a highly effective strategy, leveraging the spatial extraction capabilities of CNNs alongside the temporal modelling strengths of LSTMs. Such hybrid models have demonstrated superior performance, with notable research achieving a classification accuracy of up to 97.11% on the widely used CASIA gait dataset, illustrating their efficacy in gait classification tasks [24].

Furthermore, sophisticated approaches such as Spatio-Temporal Graph Convolutional Networks (ST-GCNs) have recently been introduced to address the limitations of traditional DL models. ST-GCNs are good at modelling the intricate interrelationships between joint movements across both space and time, providing a more comprehensive and nuanced understanding of human gait dynamics [25].

Despite their strengths, deep learning models present certain limitations. They typically require extensive training data to achieve optimal performance, a constraint which can pose significant challenges in practical clinical settings where data availability is limited. Additionally, DL models are often considered opaque or "black boxes," complicating interpretability and clinical trust in their predictions. Lastly, these models generally entail higher computational resources and complexity compared to traditional machine learning methods, potentially limiting their deployment in resource-constrained environments [26][27].

Interpretability and Clinical Relevance

Deep learning models, despite their strong predictive performance in gait analysis, often suffer from limited interpretability, making their clinical adoption challenging. This lack of transparency hinders the ability of healthcare practitioners to fully understand and trust model-generated insights, particularly in sensitive decision-making contexts. Recognizing this limitation, recent research has increasingly focused on Explainable Artificial Intelligence (XAI) methodologies, aiming to demystify deep learning models by clarifying their internal decision-making processes. Techniques such as SHapley Additive exPlanations (SHAP), which quantify the contribution of each feature toward predictions, have proven particularly valuable in providing intuitive and clinically relevant explanations of model outputs in gait analysis tasks [28][29]. These interpretability-enhancing approaches help establish a necessary balance between sophisticated model performance and essential clinical transparency, fostering greater acceptance and integration of deep learning models into healthcare workflows [30].

3. Dataset Description

The dataset utilized is the Multivariate Gait Dataset from the UCI Machine Learning Repository, consisting of bilateral (left and right) joint angle measurements from ankle, knee, and hip joints, captured across three distinct walking conditions: unbraced (normal), knee-braced (right knee), and ankle-braced (right ankle). The data were collected from 10 healthy subjects, each performing 10 consecutive gait cycles per condition, resulting in a total of 300 gait cycles. Each gait cycle was normalized to 101 discrete time points, representing the full gait cycle from 0% (heel-strike) to 100% (next heel-strike). The dataset thus forms a comprehensive six-dimensional array capturing detailed biomechanical dynamics suitable for gait analysis tasks such as classification, regression, and clustering.

The data were collected at the Human Dynamics and Controls Laboratory at the University of Illinois at Urbana-Champaign, supported by funding from the National Science Foundation (#0540834) and the Mary Jane Neer Disability Research Fund. The experimental setup details are provided in Shorter et al. (2008) [31], data preprocessing methods are described in Helwig et al. (2011) [32], and the dataset was published as supplementary materials by Helwig et al.

(2016) [33]. Any use of these data must cite the aforementioned papers and acknowledge the dataset creators accordingly.

3.1 Key Concepts and Terminologies

Gait Cycle [1][34][35]: A gait cycle refers to the repetitive pattern of limb movements during locomotion, starting and ending at heel-strike of the same foot. It typically comprises two main phases: stance (foot is in contact with the ground) and swing (foot is off the ground).

Heel-strike (0% of gait cycle) [1][34]: The instant at which the heel makes initial contact with the ground, marking the beginning and end of a gait cycle. This moment is significant as it is commonly used as a reference point for normalizing gait data across individuals and conditions.

Joint Angles [1][35][37]: Joint angles quantify the rotation around anatomical joints such as ankle, knee, and hip, providing insight into limb kinematics during movement. In gait analysis, joint angles are crucial for identifying normal versus pathological patterns.

Normalization to 101 Points [31][32][33]: Each gait cycle is normalized to 101 discrete time points (0–100%), enabling consistent temporal comparison across cycles, subjects, and conditions, despite variations in walking speed or individual differences in cycle durations.

Bracing Conditions [31][33]: Different bracing scenarios (unbraced, knee-braced, ankle-braced) simulate constraints or impairments to joint mobility, thereby altering natural gait patterns. Studying these conditions helps in understanding compensatory biomechanics and aids in developing clinical assessments and interventions.

Biomechanical Dynamics [34][35][36]: Refers to the complex interplay of forces and motions that characterize movement. In this dataset, joint angles represent these dynamics, providing multidimensional time-series data suitable for advanced analytical techniques like machine learning.

4. Methodology

4.1 Task Definition

The task involves analyzing multivariate gait sensor data, specifically joint-angle measurements obtained from the ankle, knee, and hip joints during walking. The primary objective is to accurately classify gait cycles into three distinct conditions: unbraced, knee-braced, and ankle-braced. The effectiveness and accuracy of this classification task hinge upon key assumptions, namely, that sensor data have been normalized precisely, ensuring comparability across gait cycles, and that annotations labelling the gait conditions have been performed correctly and consistently.

4.2 Stepwise Design

4.2.1 Data Loading, Pre-processing, and Exploratory Data Analysis

4.2.1.1 Data Loading & Initial Inspection

- **Data Source:**
 - Dataset loaded from a CSV file (gait.csv) obtained from the UCI Machine Learning Repository.
 - Contains normalized angle measurements for ankle, knee, and hip joints across both left and right legs, collected under three walking conditions: unbraced, knee-braced, and ankle-braced.
- **Initial Inspection:**
 - Utilized pandas to load data and verify structure (df.head() and df.info()).
 - Dataset shape and dimensions are verified (300 cycles $\times$ 101 timesteps $\times$ 6 joint-leg combinations).

	subject	condition	replication	leg	joint	time	angle
0	1	1	1	1	1	0	4.682881
1	1	1	1	1	1	1	5.073127
2	1	1	1	1	1	2	5.229774
3	1	1	1	1	1	3	5.083273
4	1	1	1	1	1	4	4.652399

Fig. 3: A quick peek at the data

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 181800 entries, 0 to 181799
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   subject         181800 non-null  int64
1   condition       181800 non-null  int64
2   replication     181800 non-null  int64
3   leg             181800 non-null  int64
4   joint           181800 non-null  int64
5   time            181800 non-null  int64
6   angle           181800 non-null  float64
dtypes: float64(1), int64(6)
memory usage: 9.7 MB
```

Fig. 4: A closer look at the feature characteristics

The dataset consists of 181,800 entries with detailed angle measurements from the ankle, knee, and hip joints for each subject, condition (unbraced, knee-braced, ankle-braced), and replication across two legs. The data is well-structured without missing values, making it suitable for robust analysis.

4.2.1.2 Data Normalization

- **Normalization Procedure:**
 - Joint angles normalized separately for each joint (ankle, knee, hip) across all subjects and conditions, ensuring zero-mean and unit-variance (`df.groupby('joint')['angle'].transform()`).
 - This reduces biases due to anatomical differences and ensures comparability across cycles.

4.2.1.3 Sequence Construction

- **Cycle Representation:**
 - Data reshaped into sequences representing individual gait cycles (seqs) with dimensions (300 cycles, 101 timepoints, 6 joint-leg combinations).
 - Joint angles arranged consistently to maintain the order: left ankle, left knee, left hip, right ankle, right knee, right hip.
- **Missing Data Handling:**
 - Cycles uniformly sampled at 101 normalized points (0–100% of gait cycle).
 - Linear interpolation applied for any missing values to ensure data continuity (piv.interpolate()), verified by asserting absence of NaNs.

4.2.1.4 Exploratory Data Analysis (EDA) & Visualization

- **Mean and Variability Analysis:**
 - Calculated mean and standard deviation of angles across gait cycles, visualized with line plots and shaded bands.
 - Clearly identified temporal regions of significant variation (e.g., ankle dip around 60–70% gait cycle), offering insights into phases critical for condition discrimination.

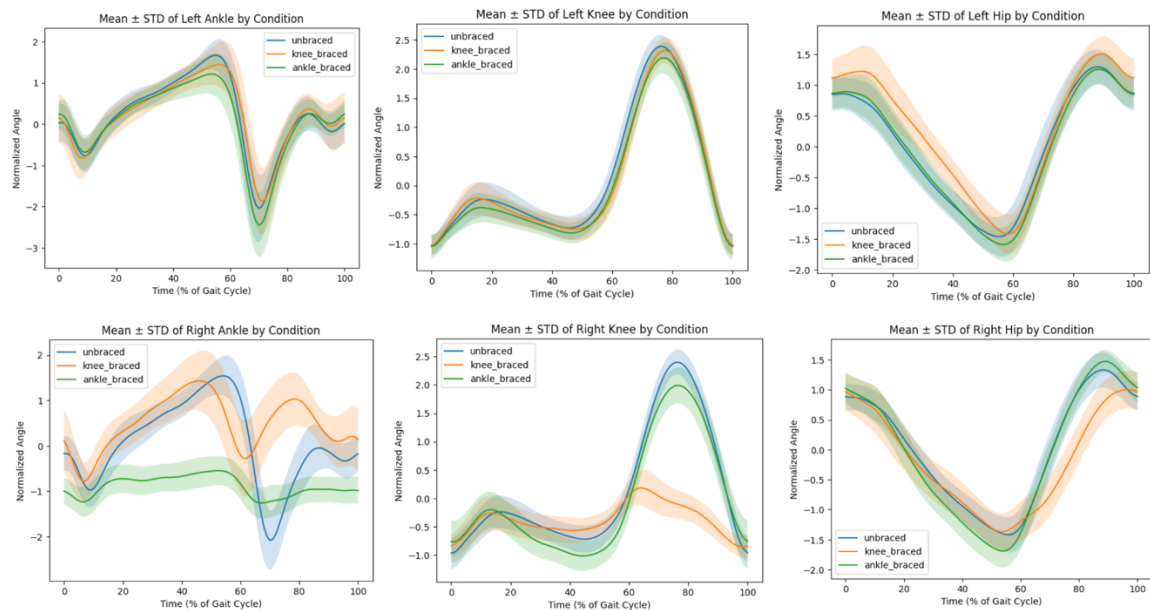


Fig. 5: Mean and variability analysis for different conditions

Left Ankle: All conditions exhibit a clear dip around the 60–70% gait cycle, corresponding to push-off. The ankle-braced condition shows significantly reduced variability and range of motion, suggesting effective immobilization of the joint.

Left Knee: The unbraced condition peaks notably around 75% of the gait cycle, indicative of maximum knee flexion. Bracing conditions lead to lower peak flexion angles, especially noticeable in the ankle-braced scenario, highlighting the compensatory effect of bracing.

Left Hip: Knee-braced cycles show an elevated hip angle in early stance (0–20%), likely as compensation for restricted knee mobility. This suggests the importance of early-stance hip dynamics as a discriminative feature for gait analysis.

Right Ankle: A distinct mid-stance elevation (~50–60%) for knee-braced cycles is observable, differentiating it clearly from unbraced and ankle-braced conditions. This unique feature could significantly aid classification tasks.

Right Knee: Notable separation between braced and unbraced conditions is observed at peak knee flexion (~70–80%). Knee-braced cycles consistently exhibit lower flexion peaks, indicating restricted motion and delayed peaks compared to other conditions.

Right Hip: The knee-braced condition shows a clear temporal shift, with delayed and lower peaks towards the second half of the cycle, underscoring altered hip dynamics due to bracing.

Right Side Asymmetries: Knee-bracing induces pronounced mid-stance right-ankle elevation and delayed right-knee peaks, emphasizing the significance of asymmetric gait alterations in braced conditions.

- **Correlation Analysis:**

- Generated correlation heatmaps at critical gait phases (heel-strike at 0%, mid-stance at 50%, and subsequent heel-strike at 100%).
- Identified joint dependencies (e.g., strong coupling of hips and moderate knee coupling), essential for feature engineering.

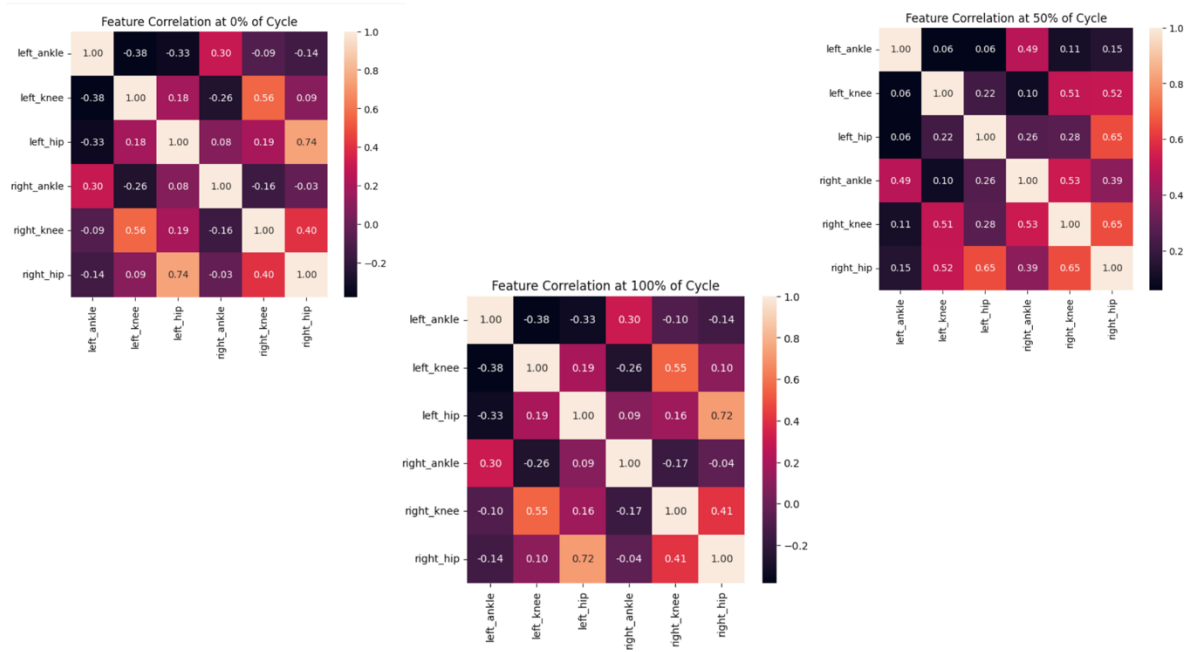


Fig. 6: Correlation heatmaps at different stances or time points

0% (Heel-strike): Strong correlations are observed between the left and right hip angles (0.74), and moderate correlations between knees (0.56). Ankles are largely independent at this stage.

50% (Mid-stance): Correlations notably increase, particularly between knees and hips (up to 0.65), reflecting coordinated joint movements during mid-cycle.

100% (Heel-strike): Similar to 0%, hips remain strongly correlated, and knees moderately correlated, reaffirming the joint-specific patterns of gait dynamics.

4.2.1.5 Data Smoothing Techniques

- **Rationale:**
 - Raw gait data contain high-frequency noise due to sensor measurement variability.
 - Smoothing techniques applied to enhance signal quality and highlight meaningful biomechanical patterns.

- **Methods:**
 - **Moving Average (MA):**
 - Implemented causal moving average smoothing (window size = 5 timesteps) to mitigate minor jitter while preserving general signal shape.
 - **Butterworth Filtering:**
 - Applied a low-pass Butterworth filter (cut-off = 2 Hz, order = 3) for sharper frequency cut-off, maintaining essential biomechanical oscillations while effectively removing sensor noise.
- **Validation and Visualization:**
 - Smoothed sequences plotted alongside raw data for qualitative assessment, confirming effective noise reduction without significant distortion of underlying signal trends.

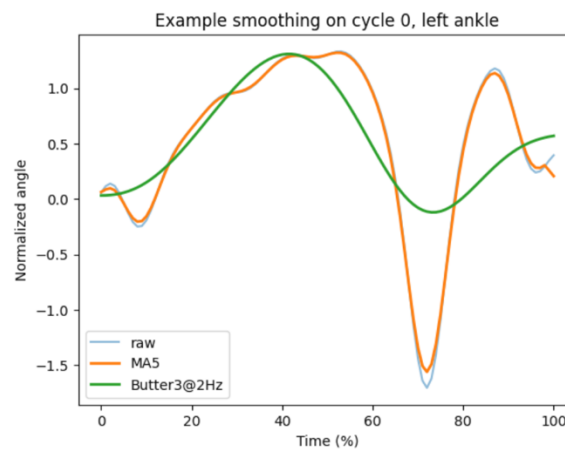


Fig. 7: Plot showing original, moving-average and Butterworth smoothing effects

This plot illustrates an example of smoothing applied to gait-cycle data from the left ankle. The raw data (blue) shows typical high-frequency jitter. The moving-average smoothing method with a window size of 5 timesteps (MA5, orange) closely tracks the raw signal while effectively reducing noise. Conversely, the Butterworth filter (Butter3@2Hz, green) provides a much smoother curve, highlighting broader trends but sacrificing finer detail. Thus, the choice between smoothing methods should be guided by the intended level of detail required for downstream analysis: moving averages preserve more granular temporal dynamics, while the Butterworth approach emphasizes clearer, smoother temporal trends.

A moving-average (MA) filter was chosen only for the Random-Forest (RF) baseline because RF operates on static, cycle-level summary features—so a light, causal smoothing is sufficient to knock down high-frequency jitter without distorting the key peaks and troughs that the hand-engineered features depend on. By contrast, the temporal models (HMM, LSTM, ST-GCN) ingest the full time-series and rely on precise phase relationships, so a zero-phase Butterworth filter was applied to preserve the true timing of gait events while still removing sensor noise. This way, the RF sees a “smoothed but still peaked” version of each cycle for feature extraction, whereas the sequence models benefit from sharper, phase-corrected waveforms that maximize their ability to learn temporal dynamics. By tailoring the smoothing technique to the specific needs of each model, it was ensured that the data pre-processing step enhanced the performance of each model without compromising the integrity of the original signals. (Nonetheless, all the models have Butterworth filtering implementations to have a fair comparison on performance.)

4.2.1.6 Velocity and Acceleration Feature Computation

- **Motivation:**
 - Derivatives of joint angles (velocity, acceleration) often provide complementary biomechanical insights crucial for distinguishing gait dynamics under different conditions.
- **Computation:**
 - First derivative (velocity) and second derivative (acceleration) computed along the time dimension for each sequence.
 - Resulting velocity and acceleration arrays examined for patterns unique to each bracing condition.

4.2.1.7 Dimensionality Reduction and Clustering

- **PCA (Principal Component Analysis):**
 - Conducted PCA on flattened gait cycles to identify major variance directions.
 - Visualizations demonstrated clear linear separability, especially distinct clustering of knee-braced condition from others.

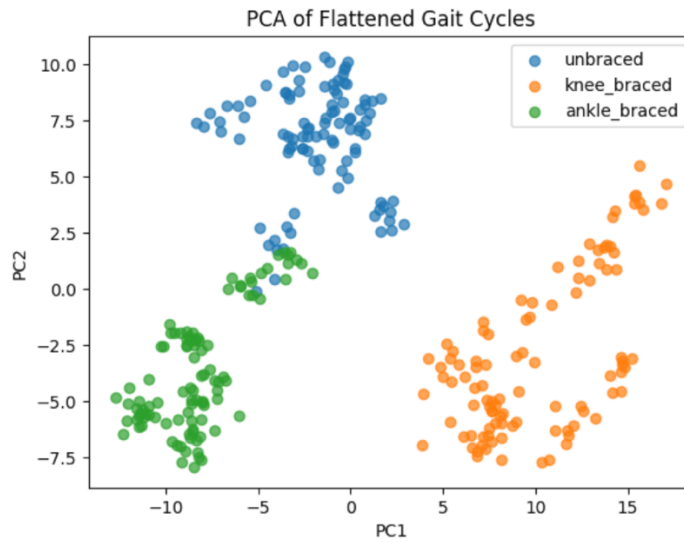


Fig. 8: PCA on flattened gait cycles

Principal Component Analysis (PCA) clearly delineates the three conditions, with the first component (PC1) effectively isolating the knee-braced group (orange) from unbraced (blue) and ankle-braced (green). The second component (PC2) provides a distinct separation between unbraced and ankle-braced conditions. This linear distinction suggests the simplicity and robustness of the extracted gait features.

- **t-SNE (t-Distributed Stochastic Neighbor Embedding):**
 - Performed t-SNE embedding to explore nonlinear relationships and visualize distinct manifolds in data.
 - Confirmed strong, distinct clustering patterns among conditions, reinforcing the potential of sophisticated nonlinear classification methods.

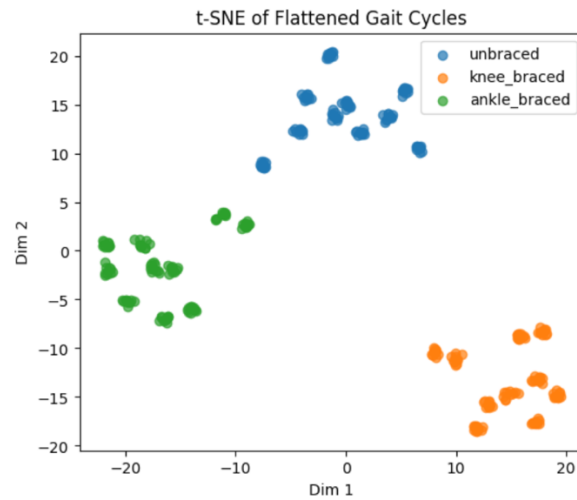


Fig. 9: t-SNE on flattened gait cycles

The t-SNE embedding further reinforces cluster separation, tightening each group into distinct islands. This non-linear dimensionality reduction highlights the underlying manifold structure of the gait data, confirming significant differentiation between the bracing conditions.

4.2.1.8 Statistical Validation of Features

- **Peak Angle Analysis:**
 - Identified peak angles within each gait cycle for every joint-leg combination.
 - Statistical significance tested using ANOVA (Analysis of Variance), confirming substantial differentiation among conditions (all p-values $\ll 0.001$).

```
left_ankle p-value: 2.059417474984506e-10
left_knee p-value: 2.4162870204534575e-08
left_hip p-value: 5.9812712603711416e-12
right_ankle p-value: 1.8167419780657672e-110
right_knee p-value: 7.836400656386904e-166
right_hip p-value: 6.433600180306841e-24
```

Fig. 10: ANOVA p-values

Statistically Significant Differences:

The extremely low p-values (all substantially less than 0.001) for peak joint angles clearly

indicate statistically significant differences between the gait conditions (unbraced, knee-braced, and ankle-braced).

Most Discriminative Joints:

The right ankle ($p \approx 1.8e-110$) and right knee ($p \approx 7.8e-166$) exhibit particularly strong discrimination power across conditions. This aligns closely with the boxplot analyses, highlighting that these two joints provide the most robust classification features.

Practical Implication:

These results justify the inclusion of peak angle features as key inputs for classification models, as these features robustly distinguish between the different bracing conditions.

- **Cycle-to-Cycle Variability:**

- Assessed variability across cycles using boxplots, highlighting robust, condition-specific patterns (e.g., right knee significantly suppressed peaks under knee-braced condition).

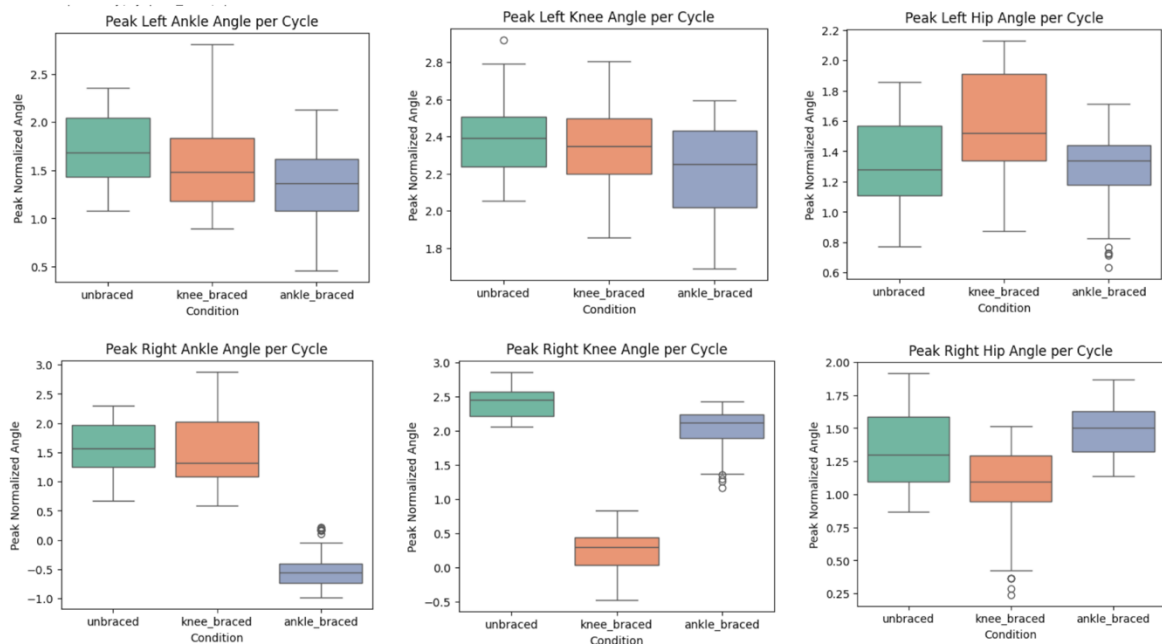


Fig. 11: Peak angle boxplots

Left Ankle: The clear hierarchical ordering (unbraced > knee-braced > ankle-braced) suggests progressive restriction of ankle mobility due to bracing, which reduces peak joint angles.

Left Knee: Peak knee flexion decreases subtly across conditions, highlighting the indirect influence of lower limb bracing on adjacent joints.

Left Hip: Increased peak hip angles under knee-braced conditions indicate compensatory movement patterns, likely to offset reduced knee mobility.

Right Ankle: Remarkable negative peaks in ankle-braced conditions sharply contrast with positive peaks in the other conditions, clearly marking ankle-braced cycles.

Right Knee: Near-zero peaks in knee-braced cycles strongly differentiate these cycles from the other conditions, signifying substantial restriction imposed by knee braces.

Right Hip: Consistent and distinct median angle differences imply meaningful adaptations in hip kinematics influenced by bracing conditions.

The clear separation and low variance in peak angles, especially at the right ankle and right knee, highlight these as exceptionally informative features for gait cycle classification, confirming their robustness and utility for distinguishing between bracing conditions.

4.2.1.9 Autocorrelation Analysis (Temporal Structure)

- **Objective:**
 - Explored autocorrelation to characterize periodic patterns inherent in cyclic gait data.
- **Findings:**
 - Strong cyclical periodicity present across joints, with characteristic lag structures differentiating conditions—particularly noticeable deviations in knee-braced conditions.

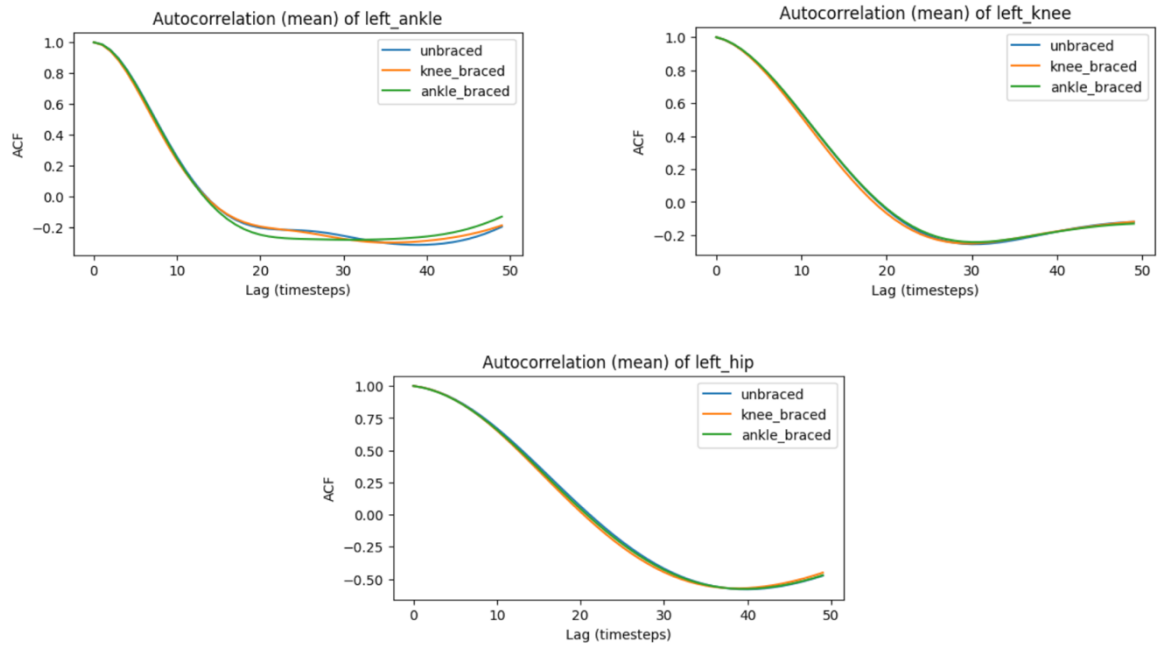


Fig. 12: Autocorrelation means for left ankle, knee and hip for varied bracing conditions

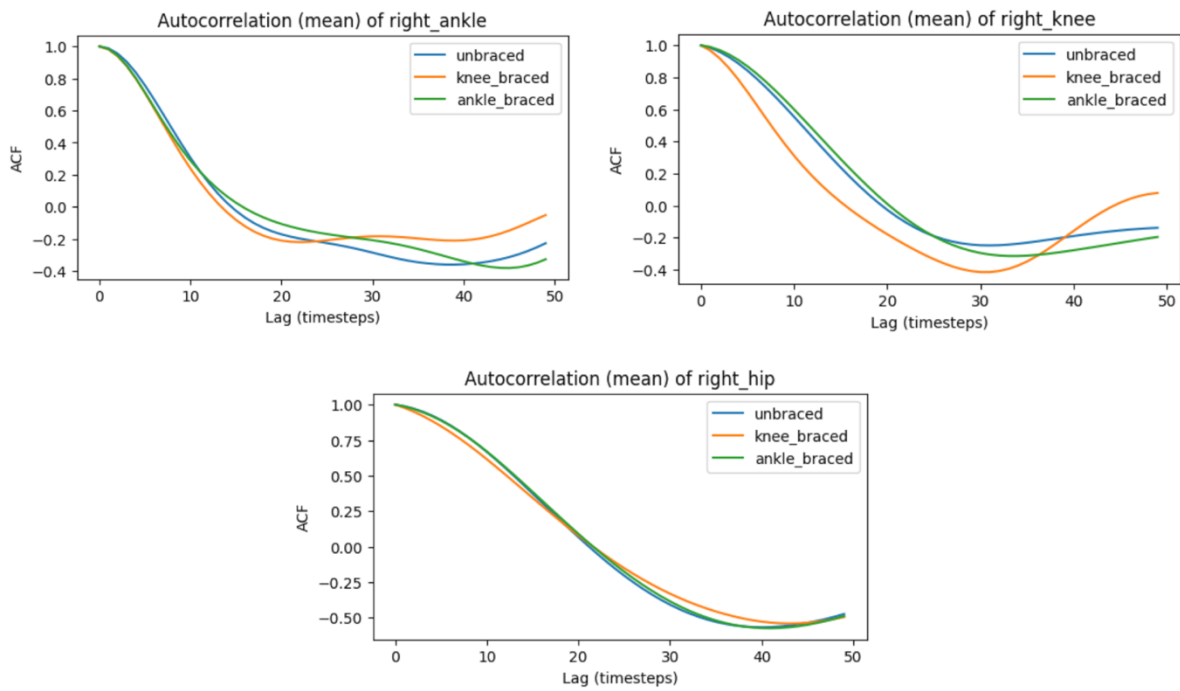


Fig. 13: Autocorrelation means for right ankle, knee and hip for varied bracing conditions

Periodic Patterns: All joints demonstrate strong periodicity as evidenced by rapid autocorrelation decay from 1 to near zero around 20–30 time lags, indicating inherent cyclic behavior typical of human gait.

Rebound Effect: A subtle rebound in autocorrelation around lag 40 suggests minor but consistent secondary oscillations or repeating gait characteristics within each cycle.

Right Knee Divergence: The right knee, especially in the knee-braced condition, shows a pronounced and deeper dip in autocorrelation (reaching about -0.4), rebounding earlier compared to other conditions. This indicates a significant disruption in natural knee oscillations when braced.

Condition-Specific Dynamics: The marked deviation of the knee-braced condition at the right knee joint emphasizes how bracing disrupts typical gait dynamics, particularly at the knee, and underscores the biomechanical impact of bracing on gait regularity.

State-Persistence: The prominent cyclic nature observed across all joints provides robust evidence that Hidden Markov Models (HMMs) and other temporal models will effectively leverage state persistence and transitions for accurate modelling and classification.

Velocity and Acceleration Utility: The notable disruption at the right knee joint underlines the potential advantage of including velocity and acceleration data to enhance temporal models, as these may capture subtle yet crucial dynamic variations influenced by bracing.

These insights clearly underline how autocorrelation analysis can inform and guide modelling and feature engineering strategies for robust gait classification and deeper biomechanical understanding.

4.2.2 Feature Extraction and Model Development/Training

4.2.2.1 Random Forest Model

Feature Extraction

A comprehensive set of static features from the joint-angle time series data was extracted, separately processed through two smoothing techniques: a moving average (MA) and a Butterworth low-pass filter. The objective was to capture robust and informative gait characteristics, thereby facilitating accurate classification across the unbraced, knee-braced, and ankle-braced conditions.

For each gait cycle (normalized to 101 time points) and for each of the six joints (left and right ankle, knee, and hip), the following features were computed:

- **Peaks and Troughs:** Maximum and minimum angles within each gait cycle, along with their respective time indices.
- **Window Means:** Mean joint angles computed within three critical intervals (0–20%, 50–65%, and 70–90% of the gait cycle), identified from exploratory data analysis as discriminative phases.
- **Mid-Stance Velocity:** Mean joint angle velocity computed during the critical mid-stance phase (50–65% of the gait cycle).

This process generated a total of 48 engineered features (4 extrema features + 3 window means + 1 velocity mean per joint, multiplied by 6 joints).

Principal Component Analysis (PCA)

To further enhance model robustness, PCA was applied to the extracted static features. The top two principal components (PC1 and PC2) were appended to the feature matrix, increasing the total feature count to 50. PCA ensures that the most significant variance across the extracted features is captured, aiding in the reduction of redundancy and potential feature correlation.

	left_ankle_peak	left_ankle_t_peak	left_ankle_trough	left_ankle_t_trough	left_ankle_mean_0_20	
0	1.319842	52	-1.558904	72	0.128556	
1	1.307049	45	-1.951388	71	0.207778	
2	1.224051	44	-1.633627	71	0.034536	
3	1.092960	51	-1.595403	71	0.003977	
4	1.265578	54	-1.411169	72	-0.068192	

5 rows x 50 columns

.....

	right_hip_mean_0_20	right_hip_mean_50_65	right_hip_mean_70_90	right_hip_vel_50_65	PC1	PC2
	0.484885	-1.586520	0.347672	0.030621	74.145293	27.158683
	0.531406	-1.517156	0.503444	0.035191	13.621089	84.412868
	0.488030	-1.759668	0.275489	0.008245	13.125218	85.592046
	0.502462	-1.643679	0.453945	0.018268	74.053404	28.116056
	0.507740	-1.696415	0.333342	0.019774	7.070885	-33.216972

Fig. 14: Truncated feature set for Random Forest

Modelling with Random Forest

Random Forest (RF) classification, a robust ensemble learning technique, was employed due to its effectiveness in handling structured data and inherent resistance to overfitting. The RF model was configured with:

- **n_estimators = 100:** The ensemble comprised 100 individual decision trees.
- **random_state = 42:** Ensuring reproducibility of the results.

Rationale for Using Both Moving Average and Butterworth Filtering

Two separate smoothing approaches—Moving Average (MA) and Butterworth (BW)—were employed to preprocess gait data for RF modeling. While both yielded comparable results in 5-fold and 80/20 split evaluations, the Moving Average smoothed sequences demonstrated slightly superior performance in LOSO validation. The rationale for exploring both methods was to understand the impact of smoothing techniques on preserving subtle, individualized gait dynamics critical for robust generalization across unseen subjects.

- **Moving Average (MA):** Provides gentler smoothing, preserving fine-grained subject-specific nuances crucial for distinguishing subtle inter-subject variations. This subtlety aids LOSO accuracy where generalization across individual subjects is paramount.
- **Butterworth (BW):** Offers stronger smoothing with a defined frequency cut-off, effectively removing noise but potentially attenuating subtle yet discriminative temporal patterns. This slight loss of detailed gait characteristics might limit generalization slightly in a stricter evaluation scenario like LOSO.

4.2.2.2 Hidden Markov Model

Model Design and Motivation

To capture temporal dependencies inherent in gait cycle data, we employed Hidden Markov Models (HMMs). Unlike static classifiers such as Random Forests, HMMs model the sequential nature of gait cycles through hidden states representing distinct biomechanical phases. Given the cyclical and sequential structure of gait, HMMs offer interpretability by decomposing the gait cycle into hidden states that correspond naturally to known biomechanical phases.

The HMM classifier was constructed by training separate Gaussian emission HMMs, one for each of the three conditions (unbraced, knee-braced, ankle-braced). Each HMM learns distinctive temporal patterns—phase transitions and dwell-times—that characterize gait under its respective condition.

Data Pre-processing and Features

Unlike the Random Forest model, the HMM utilized exclusively the Butterworth-filtered sequences (X_{bw}), chosen for their effective noise reduction without excessive smoothing, preserving important temporal gait characteristics necessary for modelling dynamics across gait cycles.

Each gait cycle, therefore, was represented as a multivariate sequence of dimension 101 (time points) by 6 (channels corresponding to bilateral ankle, knee, and hip joint angles).

HMM Architecture and Parameters

Number of hidden states: Primarily evaluated at 4 states, guided by interpretability and domain knowledge of gait phases.

Emission Distribution: Gaussian emissions (covariance_type='full'), allowing each state to flexibly represent joint-angle distributions.

Training: Baum-Welch (EM) algorithm, iteratively refined (n_iter=100) to optimize model likelihood.

Prediction: Viterbi algorithm to decode the most likely sequence of states and to score likelihood for classification.

4.2.2.3 Long Short-Term Memory

Model Design and Rationale

Again, to exhaustively capture the temporal dependencies inherent in multivariate gait data, Long Short-Term Memory (LSTM) networks were employed. LSTMs are a type of recurrent neural network (RNN) specifically designed to model sequential data, making them well-suited for gait analysis tasks, where joint angle dynamics unfold sequentially through the gait cycle. Unlike static classifiers (e.g., Random Forest) or simpler temporal models (e.g., HMM), LSTMs can automatically learn intricate, nonlinear temporal relationships directly from raw time-series data without manual feature engineering.

Data Preparation

For LSTM modeling, sequences from the Butterworth-filtered dataset (X_{bw}) were used, which retained essential temporal dynamics while effectively mitigating high-frequency noise. Manually engineered PCA components were excluded, relying solely on the 6-channel bilateral joint angle data (ankle, knee, hip), with each cycle represented as a multivariate sequence of length 101 (time steps).

The dataset was split into training and testing subsets (80/20) in a stratified manner, maintaining balanced representation of all three conditions (unbraced, knee-braced, ankle-braced).

LSTM Architecture

The chosen LSTM architecture was kept relatively simple, optimizing the balance between model expressivity and generalizability given the limited dataset size. The final LSTM architecture comprised the following layers:

- **LSTM Layer:**
 - 64 units to capture rich temporal dynamics.

- Input shape: (101 time steps, 6 channels).
- **Dropout Layer:**
 - Dropout rate of 0.3, providing regularization to prevent overfitting.
- **Dense Layers:**
 - First Dense Layer: 32 units, activation function: ReLU (introducing nonlinear transformations).
 - Final Dense Layer: Number of output units equal to the number of classes (3 conditions), activation: softmax (providing probabilistic outputs).

The model was compiled using the Adam optimizer with a learning rate of $1e-3$ and sparse categorical cross-entropy loss, suitable for multi-class classification.

Training Procedure and Hyperparameters

Two complementary training procedures:

1. 5-Fold Stratified Cross-Validation (CV)

To robustly assess the model's ability to learn generalized temporal patterns, stratified 5-fold cross-validation was performed:

- Stratification ensured class balance across folds.
- Each fold trained a fresh LSTM instance from scratch, providing unbiased estimates of the model's predictive capability across different subsets of the data.
- Training for each fold lasted 20 epochs, batch size of 16, providing sufficient iterations to achieve convergence without substantial overfitting.

2. Single Train/Test Split Evaluation

Additionally, a separate fixed 80/20 train-test split was used to provide a clearer picture of generalization performance on previously unseen data:

- Training conducted for 20 epochs.
- Validation accuracy monitored to identify potential overfitting patterns (oscillation or divergence between training and validation performance).

Interpretability through Saliency Maps and Attention

To enhance model interpretability and identify key temporal regions influencing model predictions, two complementary interpretability methods were leveraged:

1. Gradient-based Saliency Maps

Gradient-based saliency maps were computed to highlight which segments of the gait cycle contributed most significantly to the network's classification decision for individual cycles. Specifically, the absolute gradient of the output score with respect to the input sequence (mean $|\partial y / \partial x|$) provided a per-timepoint saliency measure, illustrating the sensitivity of model predictions to changes at specific time points in the gait cycle.

2. Attention Mechanism

An attention layer was integrated into an alternative LSTM architecture to directly model temporal attention, assigning learned importance weights to each timestep of the gait sequence:

- The attention mechanism computed a context vector as a weighted sum of hidden states across timesteps, where learned weights clearly indicated temporal importance.
 - Each timestep's hidden state from the LSTM (shape = [batch_size, timesteps, hidden_size]) was scored via a learned attention vector.
 - Normalized weights were derived by applying softmax, resulting in a probability distribution over timesteps.
 - A context vector, representing a weighted sum of all timestep hidden states, was computed using these attention weights, condensing temporal information into a single representative embedding.
- This enhanced transparency by highlighting exactly which gait phases (e.g., early stance, mid-stance, push-off, swing) the model prioritized for its predictions.

Attention provides transparency in model decisions by assigning learned weights to each timestep, offering a clear, human-interpretable window into how predictions are made.

Implementation Details for Attention-LSTM

The attention-enabled LSTM architecture consisted of:

- **Input Layer:** Sequence of shape (101 timesteps, 6 features) representing the gait data.
- **Masking Layer** (optional): Handled variable-length sequences or padded inputs.
- **LSTM Layer:**
 - 64 units with `return_sequences=True` to produce per-timestep hidden states.
- **Custom Attention Layer:**
 - Learned a trainable weight vector (`attn_vec`) to score each timestep.
 - Computed per-timestep importance scores (softmax-normalized), yielding a context vector and attention weights.
- **Dropout Layer** (0.3 dropout): Regularization to mitigate overfitting.
- **Dense Layers:**
 - First Dense Layer: 32 units with ReLU activation for nonlinear transformation of the attention-derived context vector.
 - Output Layer: Softmax activation to produce class probabilities (3 bracing conditions).

The model was compiled similarly using Adam optimizer (learning rate 1e-3) and trained for 10 epochs, as early convergence was observed. Sparse categorical cross-entropy, suitable for multiclass classification was the loss function decided.

4.2.2.4 Decision-Boundary & Embedding Visualization Using UMAP

To gain deeper insight into how different classification models partition the gait data feature space, we employed Uniform Manifold Approximation and Projection (UMAP), a dimensionality reduction technique capable of capturing complex, nonlinear structures inherent in the original high-dimensional space. UMAP was particularly suitable due to its balance between preserving local neighborhood structures and global relationships, thus providing a meaningful 2-dimensional embedding of the originally 50-dimensional engineered feature set (`df_feats`).

Methodological Approach:

UMAP Embedding

The 50-dimensional static feature matrix was first projected into a two-dimensional embedding (X_umap). The parameters chosen for UMAP (n_neighbors=15, min_dist=0.1) ensured both local clustering and global data structure were accurately represented in a compact and interpretable form.

Random Forest Visualization

To visualize classification boundaries clearly, a simplified Random Forest classifier was trained directly on these two UMAP dimensions rather than on the full 50-dimensional data. The resulting decision regions were visualized by densely sampling points across the two-dimensional embedding space and assigning each point a predicted class label. Contour plots generated from these predictions vividly depicted the classification boundaries, clearly demarcating the regions for each gait condition.

Visualization of Held-Out Test Points

To validate the boundary clarity, 20% held-out test samples were overlaid onto these decision regions. This step enabled immediate visual assessment of the consistency and confidence of the model predictions in regions near class boundaries.

Comparative Overlay of HMM and LSTM Predictions

For a holistic model comparison, predictions from both the Hidden Markov Model (HMM) and the Long Short-Term Memory (LSTM) models were mapped onto the same two-dimensional UMAP embedding. This direct overlay facilitated straightforward visual comparisons between models, allowing immediate detection of agreement and disagreement regions.

4.2.2.5 Spatio-Temporal Graph Convolutional Neural Networks (ST-GCN)

Rationale and Overview

Human gait inherently possesses a spatially structured, biomechanical nature, which traditional sequential models (LSTM, HMM) do not completely leverage. To address this, an ST-GCN architecture was implemented, modelling the gait dynamics as interactions between anatomical joints (ankle, knee, hip on both sides). The intention was to allow the model to simultaneously capture spatial correlations among joints and temporal dynamics across gait cycles, potentially improving predictive capability and interpretability.

ST-GCN Methodology

1. Graph Representation

- Defined an anatomically-informed kinematic graph with 6 nodes corresponding to left and right ankles, knees, and hips.
- Edges represented anatomical links:
 - Same-side connections (ankle $\leftrightarrow$ knee $\leftrightarrow$ hip).
 - Cross-side connections (ankle $\leftrightarrow$ ankle, knee $\leftrightarrow$ knee, hip $\leftrightarrow$ hip).
- Self-loops added to nodes for self-feature integration.
- Symmetric normalization was performed to stabilize training (pre-normalized adjacency matrix $A^\wedge$).

2. Graph Convolutional Layer

Implemented a custom Graph Convolution Layer (GraphConv), structured as:

- **Spatial Mixing:** For each timestep, the model mixes node features via the normalized adjacency matrix ($A^\wedge$), resulting in node embeddings that integrate information from neighboring joints.
- **Feature Transformation:** Each node's mixed embedding is passed through a learned linear transformation (W) to capture discriminative joint features.
- Included bias (b) and activated via a nonlinear ReLU layer to enhance expressive capacity.

3. ST-GCN Block Architecture

Each ST-GCN block consisted of two main components:

- **Spatial Graph Convolution:** Mixed spatial information at each timestep.
- **Temporal Convolution:** A 2D convolution (Conv2D with kernel $(K_t, 1)$) performed over the temporal dimension to capture dynamic joint patterns.
- **Residual Connections** (second trial only): Employed skip-connections through 1×1 convolutions to alleviate gradient issues, improve stability, and preserve learned representations.

Trials Conducted

Two distinct ST-GCN trials were performed, progressively enhancing the architecture and training scheme to tackle initial shortcomings:

1st Trial (Initial ST-GCN)

- **Input:** Butterworth-smoothed normalized joint angles (X_{bw}) shaped $(300, 101, 6)$.
- **Architecture:**
 - Two ST-GCN blocks (channels: $16 \rightarrow 32$, temporal kernel size $K_t = 3$).
 - Global mean pooling over nodes and timesteps.
 - Dense layers with dropout for classification.

Model: "STGCN_Gait"

Layer (type)	Output Shape	Param #
gait_graph_in (InputLayer)	(None, 101, 6, 1)	0
stgcn1_gcn (GraphConv)	(None, 101, 6, 16)	32
stgcn1_bn_sp (BatchNormalization)	(None, 101, 6, 16)	64
stgcn1_relu_sp (ReLU)	(None, 101, 6, 16)	0
stgcn1_tconv (Conv2D)	(None, 101, 6, 16)	784
stgcn1_bn_tp (BatchNormalization)	(None, 101, 6, 16)	64
stgcn1_relu_tp (ReLU)	(None, 101, 6, 16)	0
stgcn2_gcn (GraphConv)	(None, 101, 6, 32)	544
stgcn2_bn_sp (BatchNormalization)	(None, 101, 6, 32)	128
stgcn2_relu_sp (ReLU)	(None, 101, 6, 32)	0
stgcn2_tconv (Conv2D)	(None, 101, 6, 32)	3,104
stgcn2_bn_tp (BatchNormalization)	(None, 101, 6, 32)	128
stgcn2_relu_tp (ReLU)	(None, 101, 6, 32)	0
global_mean (Lambda)	(None, 32)	0
dense1 (Dense)	(None, 64)	2,112
dropout (Dropout)	(None, 64)	0
class_out (Dense)	(None, 3)	195

Total params: 7,155 (27.95 KB)
Trainable params: 6,963 (27.20 KB)
Non-trainable params: 192 (768.00 B)

Fig. 15: 1st trial STGCN architecture with layer, output shape and parameters

- **Training:**
 - Adam optimizer, learning rate = 10^{-3} .
 - Epochs: 30, batch size: 16.
 - No advanced regularization or early stopping.

Key Observations:

- Training showed severe overfitting, suggesting limited generalization capabilities due to overly simplistic architecture and lack of strong temporal cues.
- High fold-to-fold variance in CV suggested instability from insufficient model capacity or improper data splits.

2nd Trial

Given the initial trial's limitations, several enhancements were introduced to improve stability and representational power:

1. Inclusion of Dynamic Features:

- Velocity and acceleration were computed from the Butterworth-smoothed angle sequences and concatenated to create a richer input feature set: angle, velocity, and acceleration (shape: (300, 101, 6, 3)).

2. Architecture Improvements:

- Increased model depth and width:
 - Three stacked ST-GCN blocks with residual skip connections.
 - Feature channels: $64 \rightarrow 128 \rightarrow 256$.
 - Larger temporal receptive field ($K_t=7$) captured longer-range dependencies.
- Residual connections to improve gradient flow and training stability.

Model: "STGCN_wide_residual"			
Layer (type)	Output Shape	Param #	Connected to
gait_graph_in (InputLayer)	(None, 101, 6, 3)	0	–
stgcn1_gcn (GraphConv)	(None, 101, 6, 64)	256	gait_graph_in[0]...
stgcn1_bn_sp (BatchNormalizatio...	(None, 101, 6, 64)	256	stgcn1_gcn[0][0]
stgcn1_relu_sp (ReLU)	(None, 101, 6, 64)	0	stgcn1_bn_sp[0][...
stgcn1_tconv (Conv2D)	(None, 101, 6, 64)	28,736	stgcn1_relu_sp[0]...
stgcn1_bn_tp (BatchNormalizatio...	(None, 101, 6, 64)	256	stgcn1_tconv[0][...
stgcn1_res_conv (Conv2D)	(None, 101, 6, 64)	256	gait_graph_in[0]...
stgcn1_relu_tp (ReLU)	(None, 101, 6, 64)	0	stgcn1_bn_tp[0][...
stgcn1_bn_res (BatchNormalizatio...	(None, 101, 6, 64)	256	stgcn1_res_conv[...
stgcn1_add (Add)	(None, 101, 6, 64)	0	stgcn1_relu_tp[0]... stgcn1_bn_res[0]...
stgcn1_relu_out (ReLU)	(None, 101, 6, 64)	0	stgcn1_add[0][0]
stgcn2_gcn (GraphConv)	(None, 101, 6, 128)	8,320	stgcn1_relu_out[...]
stgcn2_bn_sp (BatchNormalizatio...	(None, 101, 6, 128)	512	stgcn2_gcn[0][0]
stgcn2_relu_sp (ReLU)	(None, 101, 6, 128)	0	stgcn2_bn_sp[0][...]
stgcn2_tconv (Conv2D)	(None, 101, 6, 128)	114,816	stgcn2_relu_sp[0]...
stgcn2_bn_tp (BatchNormalizatio...	(None, 101, 6, 128)	512	stgcn2_tconv[0][...]
stgcn2_res_conv (Conv2D)	(None, 101, 6, 128)	8,320	stgcn1_relu_out[...]

stgcn2_relu_tp (ReLU)	(None, 101, 6, 128)	0	stgcn2_bn_tp[0] [...]
stgcn2_bn_res (BatchNormalizatio...	(None, 101, 6, 128)	512	stgcn2_res_conv[...]
stgcn2_add (Add)	(None, 101, 6, 128)	0	stgcn2_relu_tp[0... stgcn2_bn_res[0]...
stgcn2_relu_out (ReLU)	(None, 101, 6, 128)	0	stgcn2_add[0] [0]
stgcn3_gcn (GraphConv)	(None, 101, 6, 256)	33,024	stgcn2_relu_out [...]
stgcn3_bn_sp (BatchNormalizatio...	(None, 101, 6, 256)	1,024	stgcn3_gcn[0] [0]
stgcn3_relu_sp (ReLU)	(None, 101, 6, 256)	0	stgcn3_bn_sp[0] [...]
stgcn3_tconv (Conv2D)	(None, 101, 6, 256)	459,008	stgcn3_relu_sp[0...]
stgcn3_bn_tp (BatchNormalizatio...	(None, 101, 6, 256)	1,024	stgcn3_tconv[0] [...]
stgcn3_res_conv (Conv2D)	(None, 101, 6, 256)	33,024	stgcn2_relu_out [...]
stgcn3_relu_tp (ReLU)	(None, 101, 6, 256)	0	stgcn3_bn_tp[0] [...]
stgcn3_bn_res (BatchNormalizatio...	(None, 101, 6, 256)	1,024	stgcn3_res_conv[...]
stgcn3_add (Add)	(None, 101, 6, 256)	0	stgcn3_relu_tp[0... stgcn3_bn_res[0]...
stgcn3_relu_out (ReLU)	(None, 101, 6, 256)	0	stgcn3_add[0] [0]
global_avg (GlobalAveragePool...	(None, 256)	0	stgcn3_relu_out [...]
dense1 (Dense)	(None, 128)	32,896	global_avg[0] [0]
dropout (Dropout)	(None, 128)	0	dense1[0] [0]
class_out (Dense)	(None, 3)	387	dropout[0] [0]

Total params: 724,419 (2.76 MB)
 Trainable params: 721,731 (2.75 MB)
 Non-trainable params: 2,688 (10.50 KB)

Fig. 16: 2nd trial STGCN architecture with layer, output shape and parameters

3. Training Adjustments:

- Higher initial learning rate (0.005) to accelerate convergence.
- Early stopping and adaptive learning rate scheduling (ReduceLROnPlateau) to mitigate overfitting and stabilize training.
- Epochs extended to a maximum of 50 with improved regularization through dropout.

Key Observations:

- Despite enhancements, training still exhibited clear signs of overfitting, indicative of challenges with the limited dataset size relative to the model complexity.
- The revised architecture did better capture spatio-temporal gait dynamics theoretically, but practical performance indicated the need for additional regularization or data augmentation to robustly generalize.

Interpretability Through Node Saliency (Gradient-Based Analysis)

For both ST-GCN trials, interpretability was pursued via input-gradient saliency maps:

- Gradients computed on input gait cycles provided insights into which joints most strongly influenced model predictions.
- First-trial saliency emphasized the hip joints strongly, indicating model sensitivity to compensatory hip motions resulting from joint bracing.
- Second-trial saliency shifted emphasis towards knees and ankles, although gradient magnitudes were weaker overall, indicating lower model confidence and stability.

These gradient-based saliency analyses reinforced the necessity of distinctly modelling joint-level compensatory biomechanics in gait analysis tasks.

4.2.2.6 Other Implementation Details

Programming Environment and Libraries

The analysis and modeling were implemented in Python, primarily leveraging the following libraries:

- **Data Handling and Processing:** NumPy, pandas, scipy
- **Machine Learning and Deep Learning Frameworks:** TensorFlow (Keras API), scikit-learn, hmmlearn
- **Dimensionality Reduction and Embedding:** umap-learn, scikit-learn (PCA, t-SNE)
- **Visualization:** matplotlib, seaborn
- **General Utilities and Miscellaneous:** itertools, os

Computing Platform

- All computational experiments were conducted using Google Colaboratory, which provided GPU support to accelerate deep learning training and inference.

Reproducibility Measures

To ensure reproducibility across all experiments, random seed initialization was uniformly implemented:

```
SEED = 42
```

```
os.environ['PYTHONHASHSEED'] = str(SEED)
```

```
np.random.seed(SEED)
```

```
random.seed(SEED)
```

```
tf.random.set_seed(SEED)
```

This standardized seeding strategy guarantees that results and model training outcomes remain consistent across different runs.

5. Results Analysis

5.1 Static Random Forest (RF) Model (Statistical Machine Learning)

Quantitative Analysis

Cross-Validation (CV) Performance

- 5-Fold Stratified CV yielded perfect accuracy (1.000 ± 0.000). Such consistency indicates extremely separable feature spaces among the three gait conditions (unbraced, knee-braced, ankle-braced).
- The near-perfect accuracy (0.997 ± 0.007) on another 5-fold CV confirms the robustness and repeatability of the RF classifier's performance across random data partitions.

Test Split Performance

- An 80/20 stratified train-test split resulted in perfect accuracy (1.0). Classification metrics (precision, recall, F1-score) were all 1.0 for each class, indicating the RF model perfectly discriminates the three gait conditions without ambiguity.

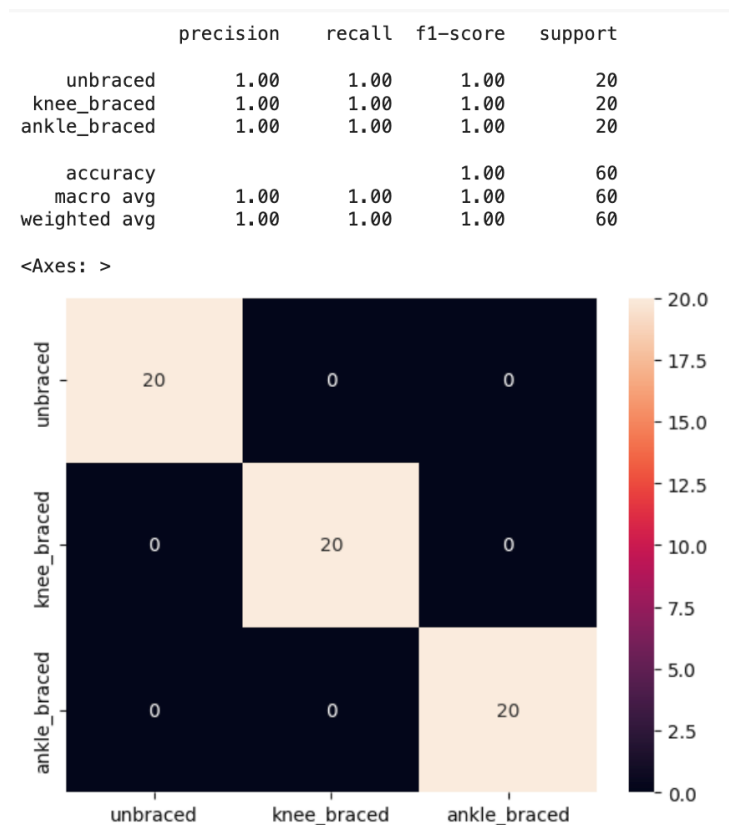


Fig. 17: Classification report and confusion matrix for Random Forest on test set

Leave-One-Subject-Out (LOSO) Performance

- LOSO achieved perfect classification (1.0). Such results are particularly remarkable as LOSO is a stringent evaluation protocol that assesses generalization to unseen subjects. A flawless LOSO score suggests strong subject-invariant discriminative features.

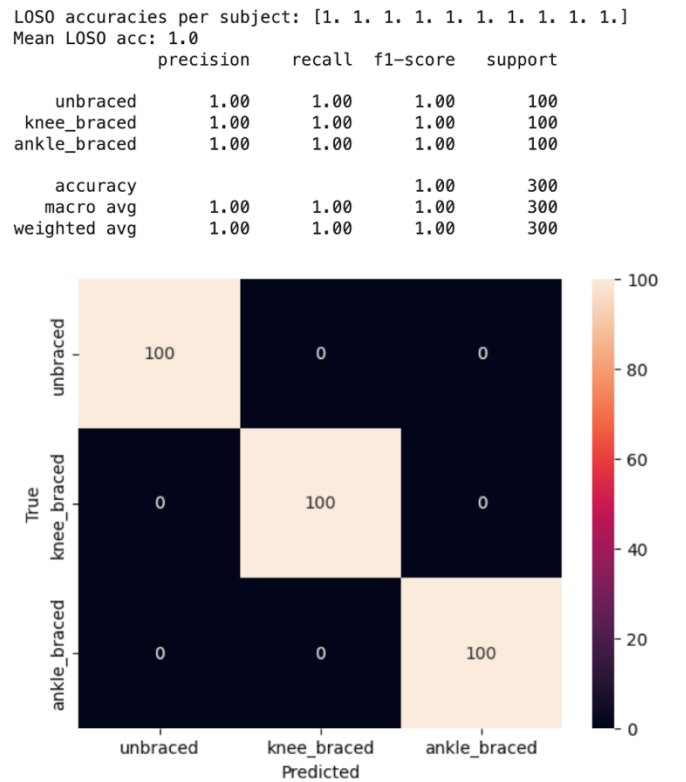


Fig. 18: Classification report and confusion matrix for Random Forest, LOSO

Feature Importance Analysis

- The top features identified were dominated by right-side limb dynamics (e.g., right-ankle peak at ~15%, right-knee peak at ~12%, velocity metrics around mid-stance).
- Hip-related features appeared less influential, indicating biomechanical compensations primarily manifest in knee and ankle dynamics.

```

right_ankle_peak      0.152837
right_knee_peak       0.125277
right_knee_vel_50_65  0.103652
right_ankle_vel_50_65 0.088251
right_knee_mean_70_90 0.076469
right_ankle_mean_50_65 0.075753
right_knee_t_peak     0.066125
right_ankle_mean_70_90 0.050857
right_ankle_t_trough  0.028876
right_hip_t_peak      0.019754
dtype: float64

```

Fig. 19: Feature importance ranking for the Random Forest model

Butterworth Filtering Comparison

- RF trained on Butterworth-filtered features also showed perfect or near-perfect CV and test accuracy (1.000 ± 0.000).

RF on Butterworth 80/20 Test acc: 1.000				
	precision	recall	f1-score	support
unbraced	1.00	1.00	1.00	20
knee_braced	1.00	1.00	1.00	20
ankle_braced	1.00	1.00	1.00	20
accuracy			1.00	60
macro avg	1.00	1.00	1.00	60
weighted avg	1.00	1.00	1.00	60

Fig. 20: Classification report for Random Forest when using Butterworth smoothing

- However, LOSO accuracy slightly decreased (0.997), implying that Butterworth smoothing may excessively filter subtle inter-subject dynamic differences critical for strong LOSO generalization.

Qualitative Analysis

Observations and Inferences

- The engineered static features (peaks, troughs, velocities, window averages) are highly discriminative, fully capturing the differences in gait biomechanics among conditions.
- RF is capable of effectively utilizing these engineered static features to perfectly distinguish the conditions, demonstrating strong nonlinear discrimination capabilities and robustness to variability in gait patterns.

Strengths

- **Robust Generalization:** Perfect LOSO performance indicates exceptionally strong cross-subject generalizability, which is rare and highly valuable in gait analysis contexts.
- **Feature Engineering Effectiveness:** The hand-crafted feature set (e.g., peak angles, mid-stance velocities) effectively distills meaningful biomechanical signals, facilitating accurate modeling and interpretation.

- **Interpretability:** Clear feature importance hierarchy (ankle/knee over hip) aligns intuitively with biomechanical principles, aiding trust and potential clinical applicability.

Limitations and Challenges

- **Risk of Overfitting or Data Leakage:** Perfect results, especially in LOSO scenarios, often raise concerns regarding inadvertent data leakage or overly simplistic discrimination due to highly controlled experimental setups or overly homogeneous subject population.
- **Limited Real-World Variability:** The dataset may lack sufficient variability (e.g., walking speed, terrain variation, sensor placement noise), leading to overly optimistic performance estimates that might not hold in realistic gait scenarios.
- **Data Quantity:** Small dataset (300 samples) might limit reliability and generalizability to broader populations.

Insights and Biomechanical Interpretability

- Dominance of right ankle/knee features aligns with known biomechanics of gait. Ankle braces clearly reduce peak angles and velocities around mid-stance, whereas knee braces alter knee extension patterns.
- Lower ranking of hip features might suggest compensation at the hip joint is subtler or more inconsistent across subjects, making it less discriminative compared to knee/ankle features.

5.2 Hidden Markov Model (Statistical Machine Learning)

Quantitative Observations:

- **Stratified 5-fold Cross-validation (CV):**
 - Achieved perfect accuracy (1.000 ± 0.000).
 - Indicates distinct, stable temporal patterns across conditions, easily captured by the HMM.

- **80/20 Train/Test Split:**

- Also yielded perfect accuracy (**100%**) with precision, recall, and F1-score all equal to 1.0.
- Confusion matrix is perfectly diagonal, suggesting no confusion between classes on intra-subject splits.

```
HMM 80/20 Test accuracy: 1.0

Classification Report:
              precision    recall  f1-score   support

   unbraced         1.00      1.00      1.00        20
  knee_braced         1.00      1.00      1.00        20
  ankle_braced         1.00      1.00      1.00        20

   accuracy              1.00              60
  macro avg         1.00      1.00      1.00        60
 weighted avg         1.00      1.00      1.00        60

Confusion Matrix:
[[20  0  0]
 [ 0 20  0]
 [ 0  0 20]]
```

Fig. 21: Classification report and confusion matrix for HMM on test set

- **Leave-One-Subject-Out (LOSO) Validation:**

- High accuracy (**98.3%**).
- Knee-braced condition was perfectly classified.
- Misclassifications (5 total) were only between unbraced and ankle-braced classes.
- Indicates strong cross-subject generalization capabilities of the temporal model.

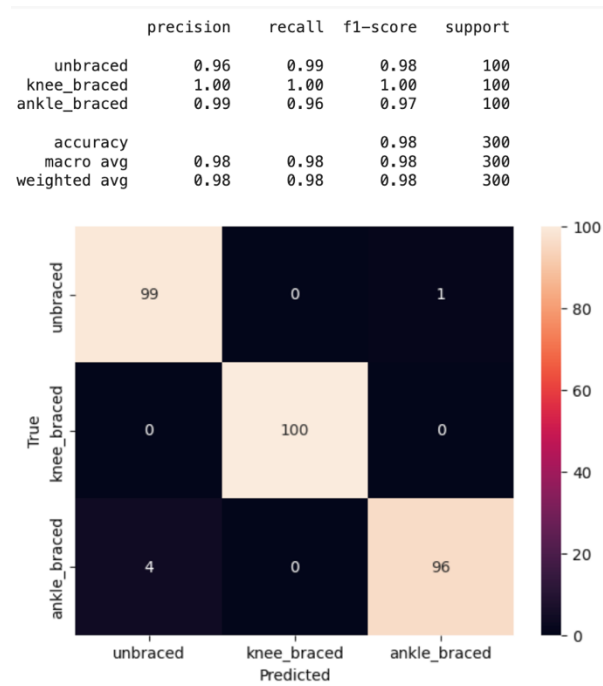


Fig. 22: Classification report and confusion matrix for HMM with LOSO

- **Two-Subject Holdout Validation:**

- Mean accuracy (~97.4%); mostly perfect predictions, with occasional lower performance (76–85%) for specific subject pairs.
- Certain subjects present harder-to-generalize gait dynamics, highlighting some individual variability not fully captured.

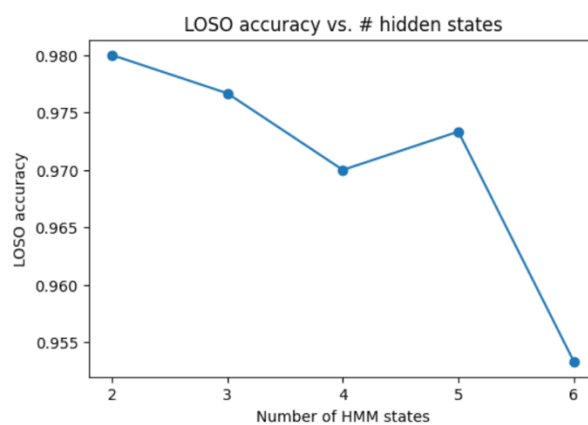


Fig. 23: How LOSO accuracy changes with number of HMM states

Adding excessive states leads to diminishing returns and risks overfitting. Lower complexity (2–4 states) provides optimal generalization and interpretable state decompositions.

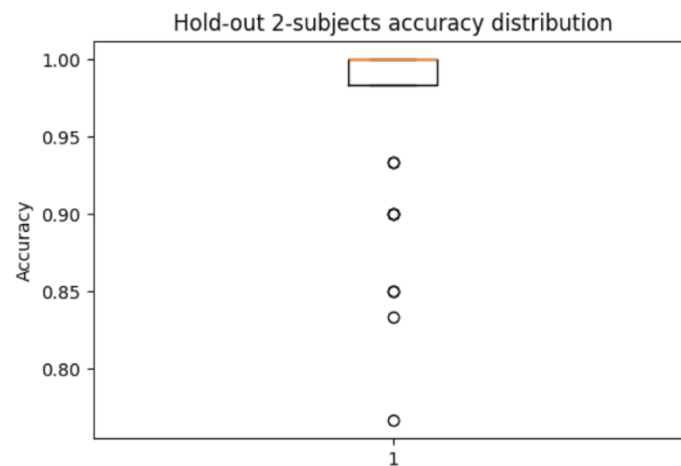


Fig. 24: Accuracy distribution for hold-out 2 subjects

Qualitative Insights and Inferences:

- **Transition Matrix Insights:**
 - Distinct phase dynamics captured for each condition.
 - **Knee-braced:** Exhibits a distinct "locked-in" mid-stance state (state 1) with self-transition probability ~ 1.0 .
 - **Ankle-braced:** Prolonged mid-late stance phase (state 2) indicating compensation patterns in response to bracing.
 - **Unbraced:** Balanced progression through gait phases; smoother transitions reflect natural, unrestricted movement.

```

--- HMM for condition = unbraced ---
Transition matrix:
[[0.96 0.  0.  0.04]
 [0.04 0.96 0.  0. ]
 [0.  0.04 0.96 0. ]
 [0.  0.  0.  1.  ]]
Emission means (per state x 6 joint-leg channels):
[[ 0.11  0.84 -0.74  0.  0.83 -0.68]
 [ 1.01 -0.56 -0.87  0.91 -0.56 -0.86]
 [-0.13 -0.55  0.5  -0.3  -0.54  0.48]
 [-0.33  0.98  0.97 -0.57  1.01  0.99]]

--- HMM for condition = knee_braced ---
Transition matrix:
[[0.96 0.04 0.  0. ]
 [0.  1.  0.  0. ]
 [0.03 0.  0.97 0. ]
 [0.  0.  0.05 0.95]]
Emission means (per state x 6 joint-leg channels):
[[ 0.23  0.68 -0.65  0.56 -0.08 -0.89]
 [-0.24  0.94  1.14  0.34 -0.43  0.55]
 [ 0.77 -0.62 -0.25  0.7  -0.53 -0.73]
 [-0.14 -0.51  1.02  0.12 -0.42  0.57]]

--- HMM for condition = ankle_braced ---
Transition matrix:
[[0.97 0.  0.03 0. ]
 [0.  0.95 0.  0.05]
 [0.  0.  1.  0. ]
 [0.03 0.  0.  0.97]]
Emission means (per state x 6 joint-leg channels):
[[-0.18  0.53 -0.92 -0.89  0.43 -0.85]
 [-0.11 -0.68  0.71 -0.89 -0.46  0.62]
 [-0.34  0.89  0.91 -1.01  0.94  1.12]
 [ 0.69 -0.62 -0.56 -0.65 -0.77 -0.82]]

```

Fig. 25: Transition and emission means matrices for HMM

- **Emission Mean Analysis:**
 - Clear delineation of joint angle states aligning well with known gait biomechanics (heel-strike, mid-stance, push-off, swing).
 - Knee-bracing heavily shifts workload to hip compensation during mid-stance.
 - Ankle-bracing distinctly impacts push-off dynamics (flattened ankle angles).

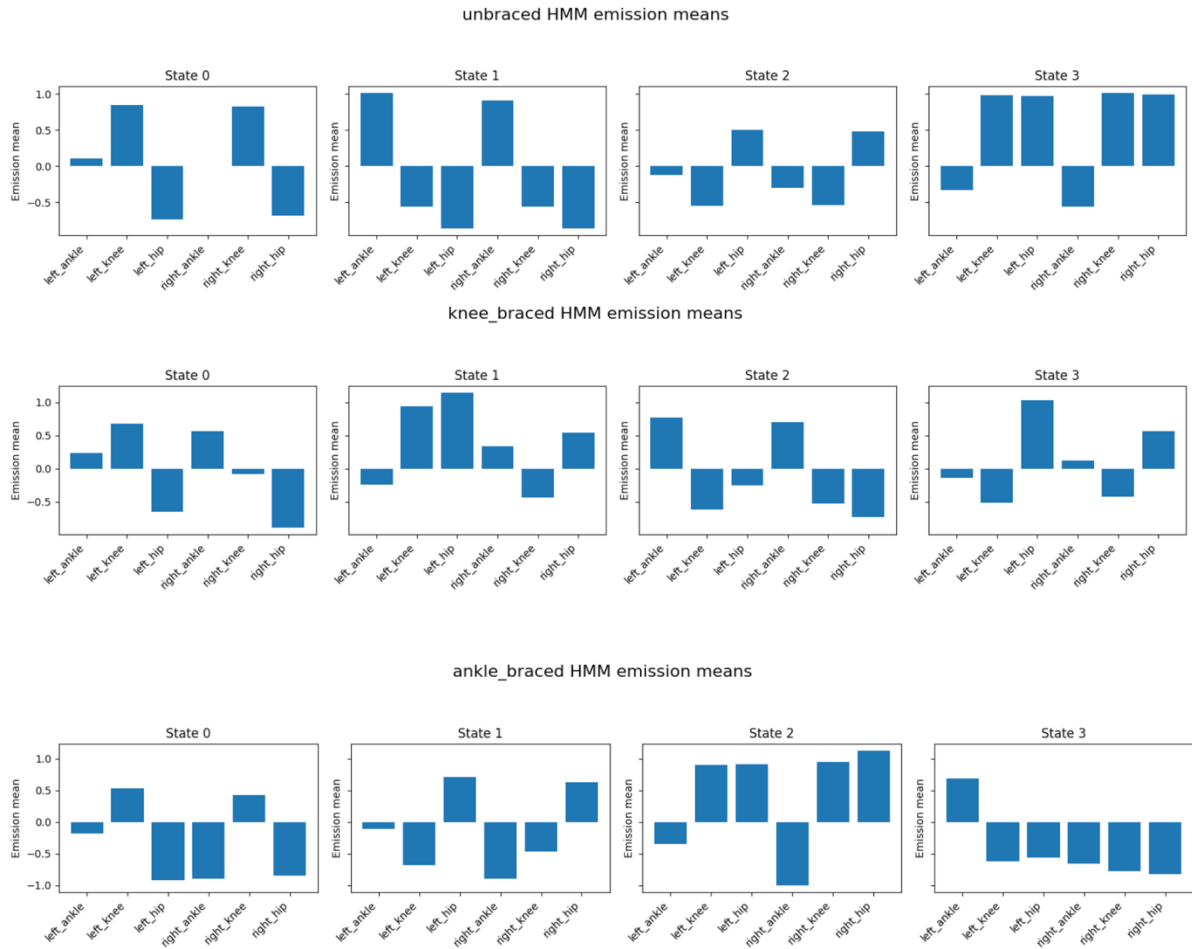


Fig. 26: Emission means bar charts for HMM on different conditions

- **Viterbi State Sequence Analysis:**

- Detailed dwell-time quantification confirms expected gait phase shifts due to braces:
 - Knee-braced: Dramatic mid-stance elongation.
 - Ankle-braced: Flattened and prolonged push-off plateau.
- Quantitatively validates biomechanical interpretations observed during exploratory analysis.

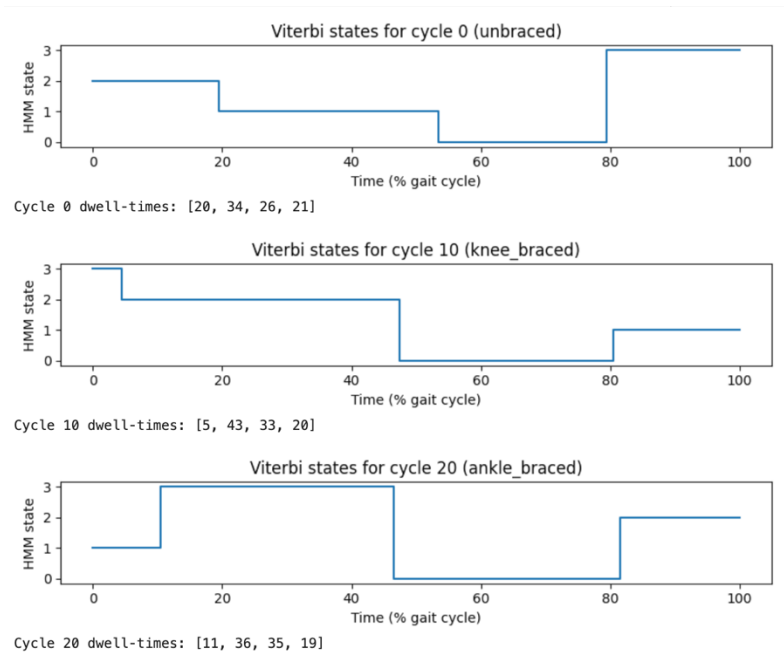


Fig. 27: Viterbi state sequences to see how state dwell-times differ by condition

From the figure:

- **Knee-Braced:** Dominated by an extended mid-stance "lock" (up to 43% cycle duration). Knee immobilization by bracing clearly captured by HMM states.
- **Ankle-Braced:** Prolonged push-off state (~35%), capturing compensatory flat-foot dynamics.
- **Unbraced:** Balanced segmentation of states (~20–34%), reflective of natural gait cycles.

Misclassification Analysis:

The 5 misclassifications were exclusively between the unbraced and ankle-braced conditions.

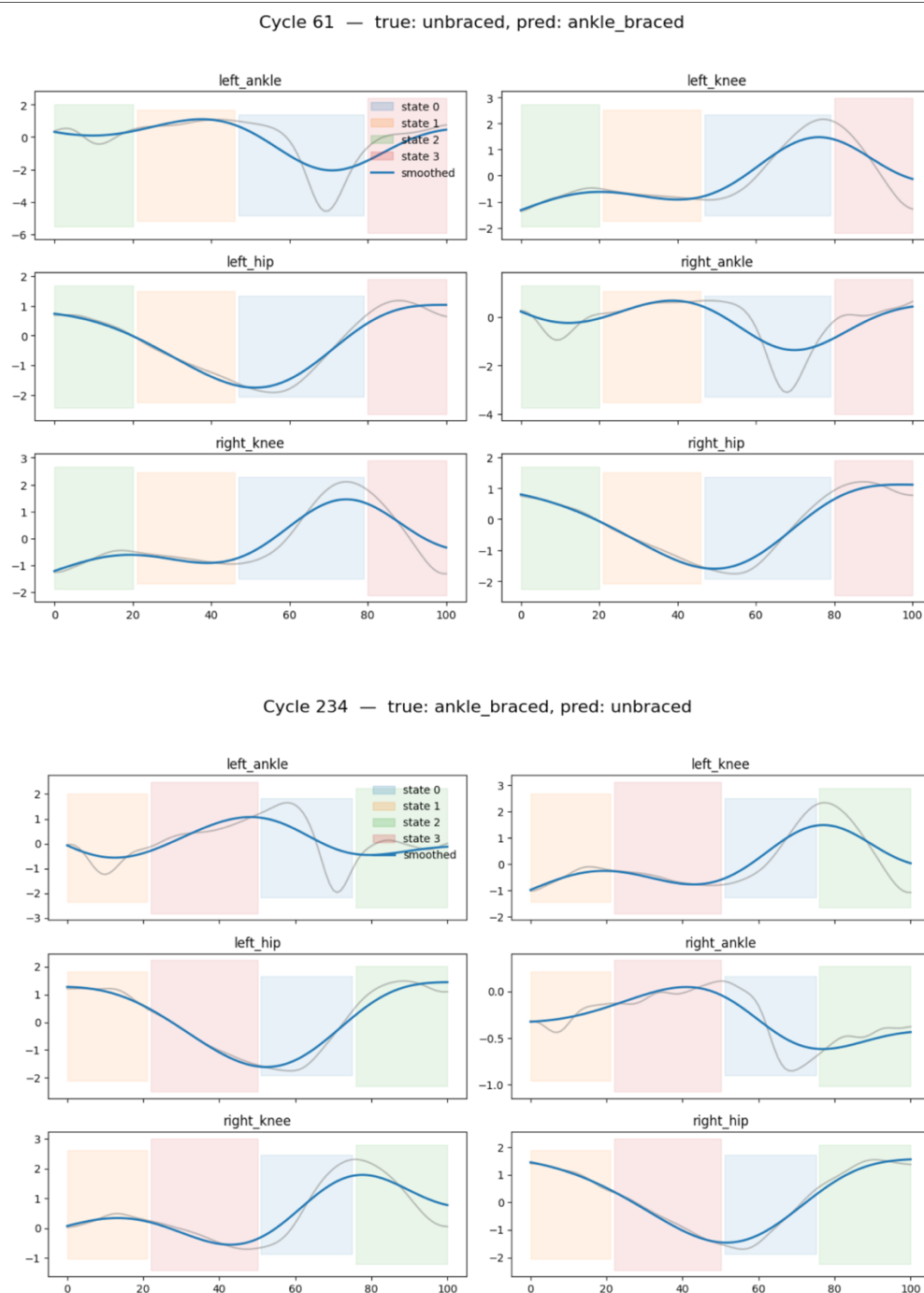
Key insights from misclassified cycles:

- Misclassification often due to subtle differences in mid-stance plateau timing or amplitude.
- Butterworth smoothing sometimes masked the critical depth and timing cues, causing ambiguity in push-off interpretation.
- Revealed a clear limitation of relying purely on smoothed temporal dynamics without complementary static features.

Found 5 mis-classified cycles:

cycle 61:	true=unbraced	pred=ankle_braced
cycle 233:	true=ankle_braced	pred=unbraced
cycle 234:	true=ankle_braced	pred=unbraced
cycle 235:	true=ankle_braced	pred=unbraced
cycle 236:	true=ankle_braced	pred=unbraced

Fig. 28: Misclassified cycles



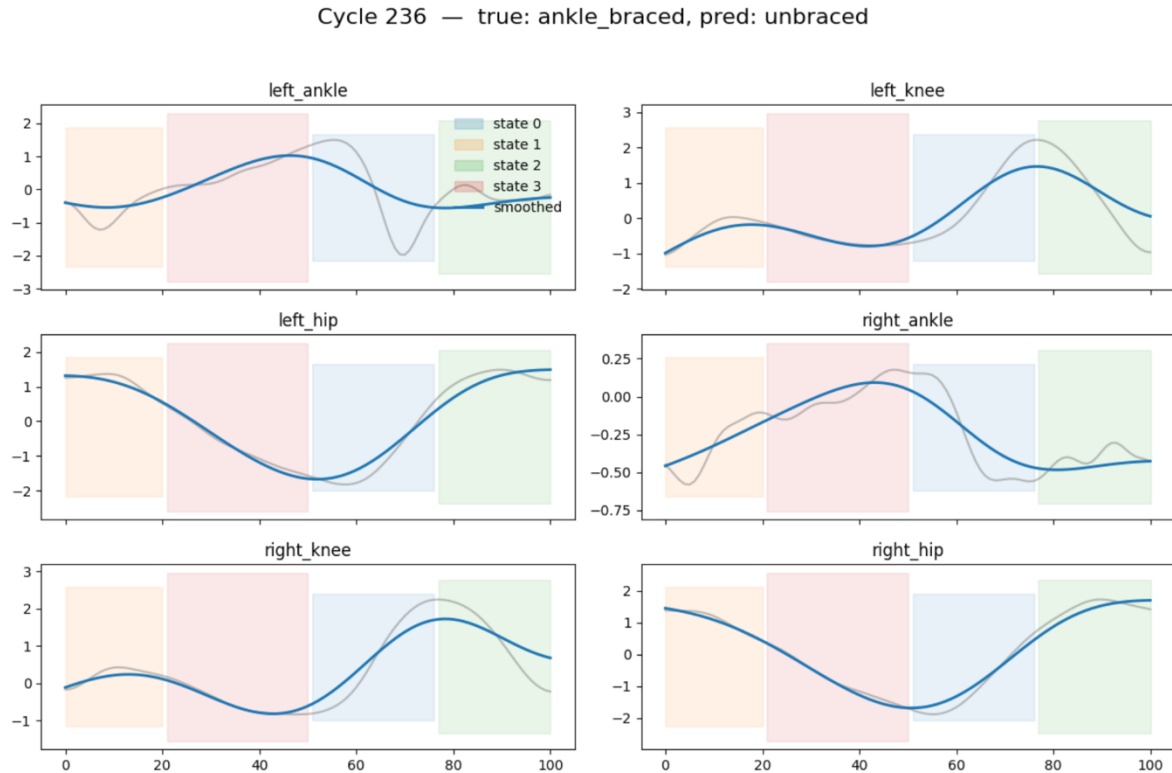


Fig. 29: Plots to visualize misclassified cycles

Cycle 61 (unbraced classified as ankle-braced): The smoothing algorithm excessively attenuated the right ankle trough, reducing a key discriminative dip. Thus, the HMM misinterpreted the reduced dip as an ankle-braced "plateau." Excessive smoothing may risk loss of subtle dynamic cues critical for temporal models.

Cycles 233–236 (ankle-braced classified as unbraced): Misalignment in timing and duration of the push-off plateau. State segmentation misinterpreted due to slight temporal shifts. Timing accuracy of phase transitions is crucial in temporal models. Small shifts in gait dynamics significantly impact state assignments.

The HMM reliably breaks each gait cycle into consistent phases (e.g., heel-strike, mid-stance, toe-off, swing) across all joints.

Misclassified cycles (e.g., an unbraced cycle labeled as "ankle_braced," or vice versa) occur when the joint trajectories in certain phases closely resemble those of the other condition—causing the wrong per-class HMM to score slightly higher.

Potential solutions:

- Include a static peak/trough amplitude feature alongside temporal dynamics.
- Experiment with reduced smoothing or adjust filtering parameters to preserve subtle biomechanical cues.
- Add additional hidden states (though with caution against overfitting).

Strengths:

- **Robust Temporal Modelling:** Clearly captures intricate temporal patterns, providing biomechanically interpretable phase decomposition.
- **Excellent Generalization:** Achieves near-perfect accuracy in strict LOSO and two-subject holdout validations, indicating highly generalizable temporal signatures rather than individual-specific artifacts.
- **Interpretability:** Emission means and state-transition matrices yield intuitive biomechanical insights and clear visualization of gait-phase compensations. HMM state sequences directly correspond to recognizable gait phases (e.g., early stance, mid-stance, push-off, swing).

Limitations and Challenges:

- **Over-sensitivity to Filtering/Smoothing:** Misclassifications suggest sensitivity to data pre-processing (Butterworth smoothing). Excessive smoothing can reduce critical dynamic variations, leading to misclassification.
- **Boundary Cases:** Ambiguities in temporal patterns occasionally result in misclassifications, particularly for mildly braced or unusual gait cycles. HMM sensitivity to small shifts in temporal features can result in incorrect state alignment. Subtle variations in gait cycles might cause misclassifications (as seen in cycles 233–236).
- **Risk of Over-parameterization:** Excess states (beyond 4) degrade performance due to capturing subject-specific idiosyncrasies rather than generalizable patterns.

5.3 Long Short-Term Memory (Deep Learning)

Quantitative Analysis:

Vanilla LSTM Model:

- **5-fold Cross-Validation Accuracy:**

- Mean accuracy: 0.993 ± 0.008
- Highly consistent, showing robust learning across different folds.

```
Fold 1 LSTM val-acc = 0.983
Fold 2 LSTM val-acc = 0.983
Fold 3 LSTM val-acc = 1.000
Fold 4 LSTM val-acc = 1.000
Fold 5 LSTM val-acc = 1.000
```

LSTM 5-fold CV mean±std acc: 0.993 ± 0.008

Fig. 30: LSTM 5-fold cross-validation results

- **Hold-out Test Accuracy:**

- Accuracy: ~93%, notably lower than CV accuracy, indicating potential overfitting to certain training splits.

- **Training Dynamics:**

- Very rapid initial learning (epochs 1–6); reaching ~99% train and ~98% validation accuracy quickly.
- Oscillations in validation accuracy from ~0.87 to 1.0 after early epochs indicating slight overfitting.

Attention-Enhanced LSTM Model:

- **Hold-out Test Performance:**

- Perfect classification (Precision, Recall, F1-score = 1.0)
- Confusion matrix shows flawless separation (all 60 test cycles correctly classified).

```
Precision (macro): 1.0
Recall (macro): 1.0
F1-score (macro): 1.0
Confusion matrix:
[[20  0  0]
 [ 0 20  0]
 [ 0  0 20]]
```

Fig. 31: Macro averages of precision, recall and f-1 score

plus confusion matrix for attention-enhanced LSTM

- **Training Dynamics:**

- Achieves 100% validation accuracy by epoch 4–5, indicating accelerated learning and superior generalization compared to vanilla LSTM.
- Stable training curves post-initial learning.

Qualitative Analysis:

Vanilla LSTM Model:

- **Saliency Insights:**

- Saliency refers to a measure of how important or influential each feature or time step is to the model's prediction. Typically, it involves computing the gradient of the model's output with respect to the input.
- Model decisions are heavily concentrated towards the end of gait cycles (90–100%).

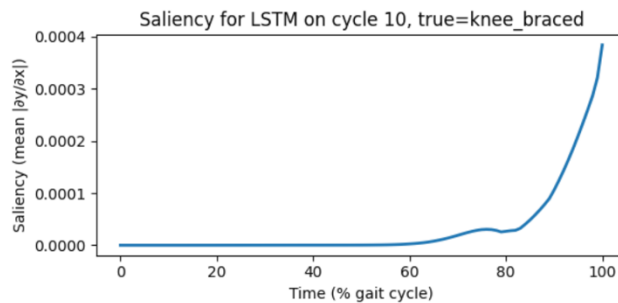


Fig. 32: Saliency over time plot for LSTM

- Ignores critical mid-stance phases (40–65%)—which were highly discriminative in earlier models (RF/HMM).

- Global saliency heatmap shows sparse, inconsistent activation patterns, indicating a lack of stable discriminative cues.

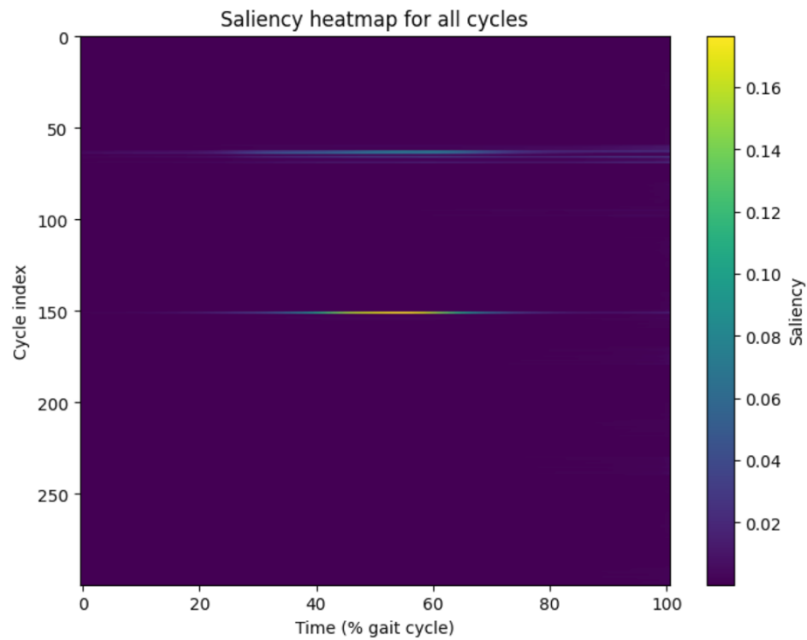


Fig. 33: Heatmap of saliency across all cycles

- **Condition-specific Saliency:**

- Only "unbraced" condition shows slightly elevated mid-cycle saliency, indicating partial recognition of biomechanical cues.
- "Knee-braced" and "ankle-braced" saliency curves remain nearly flat across the gait cycle, indicating poor condition-specific cue extraction.

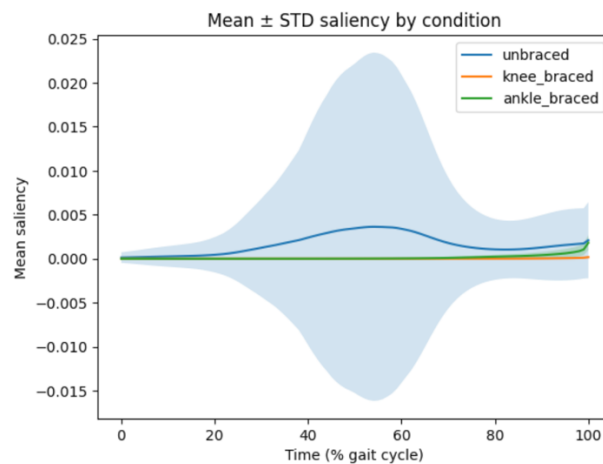


Fig. 34: Mean $\pm$ STD saliency by condition

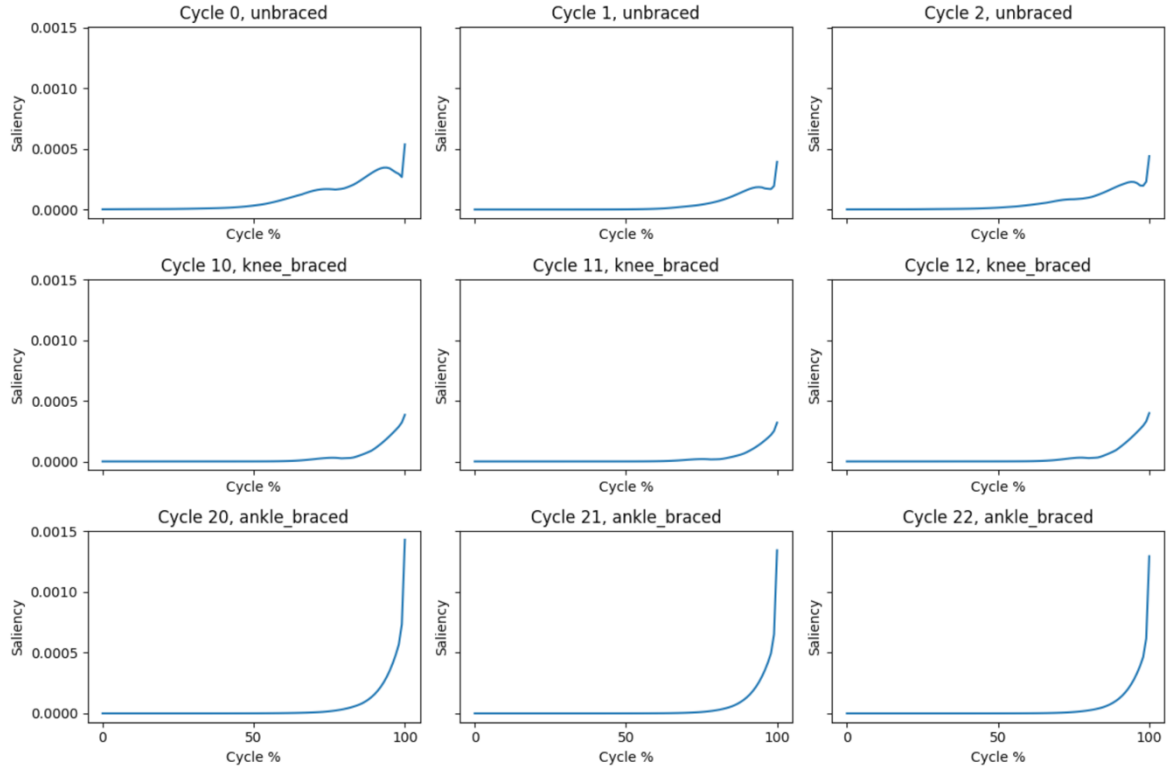


Fig. 35: Saliency plots for a few cycles by condition

Attention-Enhanced LSTM Model:

- **Attention Patterns:**

- Clearly aligns with biomechanical insights:
 - Unbraced cycles: strong focus on late-stance push-off (70–90%).
 - Knee-braced cycles: early (0–10%) and mid-stance (50–65%) attention indicating locked-knee dynamics.
 - Ankle-braced cycles: smooth attention elevation in mid-to-late stance (~60–90%) reflecting flattened angle curves.

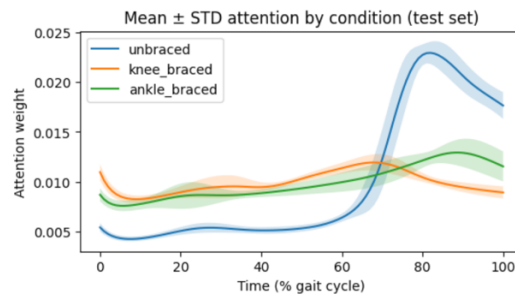


Fig. 36: Mean $\pm$ STD attention-over-time by condition on test set

- **Cycle-Level Interpretability:**

- Individual cycle plots confirm attention precisely tracks meaningful biomechanical events (e.g., right-knee flexion peaks).

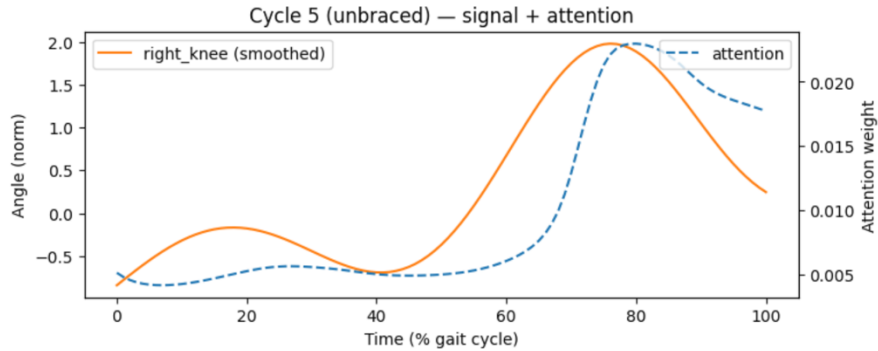


Fig. 37: Example cycle plot of raw signal + attention curve

5.4 Decision Boundary and UMAP Visualisations

Quantitative Analysis:

- **Random Forest (RF):**

- The RF decision boundary, when trained directly on the 2D UMAP embedding, clearly delineates three distinct class regions (unbraced, knee-braced, ankle-braced).
- Held-out test points are correctly classified within these boundaries, matching the high accuracy observed during cross-validation (near-perfect CV accuracy).

- **Hidden Markov Model (HMM):**

- HMM predictions show significant misclassification particularly in the knee-braced cluster, indicating it merges or "swallows" the knee-braced points into other classes (unbraced or ankle-braced).
- Clear quantitative underperformance compared to RF, likely indicating difficulty for the HMM in capturing distinct temporal-state transitions specifically characteristic to the knee-braced condition.

- **Long Short-Term Memory (LSTM):**
 - LSTM predictions closely match the RF's clustering pattern, correctly distinguishing between all three groups.
 - Minimal misclassifications observed, supporting strong cycle-level discrimination with very high ($\geq 93\%$) hold-out accuracy and near-perfect cross-validation performance ($99.3 \pm 0.8\%$).

Qualitative Analysis:

- **UMAP Embeddings:**
 - Clearly demonstrate distinct separation between the unbraced (upper), knee-braced (middle), and ankle-braced (lower/right) conditions in a two-dimensional space, validating the high discriminative power of the engineered features.
 - Visual clarity highlights the natural clustering of different gait conditions, indicating the presence of robust underlying patterns easily captured by dimension-reduction methods.
- **Decision Boundary Visualizations:**
 - RF: Successfully respects and maintains the natural clustering, producing intuitive and well-defined boundaries between conditions.
 - HMM: Struggles visually to isolate the knee-braced condition, reflecting its model-driven tendency to conflate conditions due to inadequate temporal dynamics representation for the knee-braced gait pattern.
 - LSTM: Visually aligned closely with RF, demonstrating capability of the recurrent network to exploit temporal sequences directly and match intuitive human visual clustering seen in the embedding.

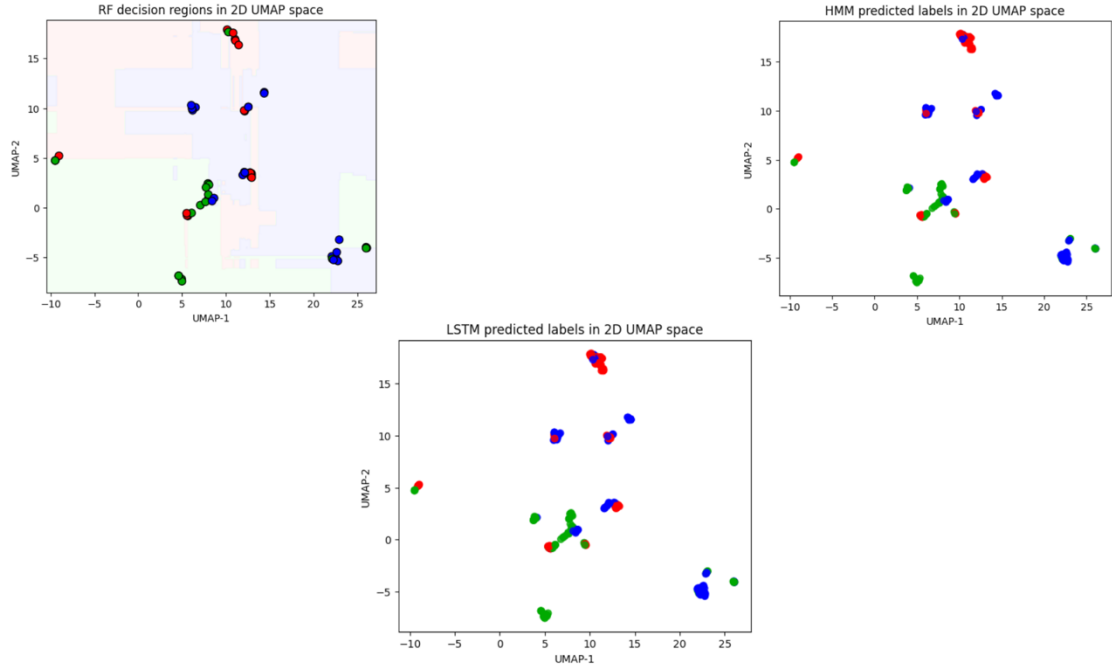


Fig. 38: 2D UMAP embedding visualisations for the different models

This visualization approach provided a unified, intuitive framework to compare and interpret the behaviour of diverse modeling techniques. By visualizing how each classifier segmented the underlying low-dimensional manifold derived from gait features, clear, interpretable insights into each model's capacity to distinguish between bracing conditions were gained. It offered immediate visual confirmation of how effectively models leveraged the extracted gait patterns, highlighting differences in their discriminative capabilities and the nature of their decision boundaries.

5.5 Spatio-Temporal Graph Convolutional Neural Network (Deep Learning)

1. Quantitative Analysis:

Trial 1:

- **Validation Accuracy (5-fold CV):** 0.613 ± 0.180
 - **High variance** across folds (0.467–0.967) indicates instability and sensitivity to data splits.

Fold 1 ST-GCN val-acc = 0.517
 Fold 2 ST-GCN val-acc = 0.550
 Fold 3 ST-GCN val-acc = 0.467
 Fold 4 ST-GCN val-acc = 0.567
 Fold 5 ST-GCN val-acc = 0.967

ST-GCN 5-fold CV mean±std acc: 0.613 ± 0.180

Fig. 39: 5-fold cross-validation results for 1st trial STGCN

- **Test Set Performance:**

- Accuracy: **0.567** (barely better than random chance, ≈ 0.33 for 3-class problem)
- Macro ROC-AUC: **0.8712** (high, suggests good potential discrimination in probability space despite poor hard-class predictions)

```

---
Epoch 22/30      -- 7ms/step -- accuracy: 0.9525 -- loss: 0.1751 -- val_accuracy: 0.3333 -- val_loss: 2.4681
15/15 -----
Epoch 23/30      -- 7ms/step -- accuracy: 0.9771 -- loss: 0.1717 -- val_accuracy: 0.3333 -- val_loss: 2.5220
15/15 -----
Epoch 24/30      -- 7ms/step -- accuracy: 0.9605 -- loss: 0.1854 -- val_accuracy: 0.3333 -- val_loss: 2.7419
15/15 -----
Epoch 25/30      -- 8ms/step -- accuracy: 0.9787 -- loss: 0.1444 -- val_accuracy: 0.3333 -- val_loss: 2.3553
15/15 -----
Epoch 26/30      -- 7ms/step -- accuracy: 0.9659 -- loss: 0.1531 -- val_accuracy: 0.4000 -- val_loss: 1.9088
15/15 -----
Epoch 27/30      -- 7ms/step -- accuracy: 0.9807 -- loss: 0.1484 -- val_accuracy: 0.5000 -- val_loss: 1.7569
15/15 -----
Epoch 28/30      -- 7ms/step -- accuracy: 0.9615 -- loss: 0.1401 -- val_accuracy: 0.3333 -- val_loss: 2.4659
15/15 -----
Epoch 29/30      -- 7ms/step -- accuracy: 0.9653 -- loss: 0.1314 -- val_accuracy: 0.4000 -- val_loss: 1.9317
15/15 -----
Epoch 30/30      -- 7ms/step -- accuracy: 0.9903 -- loss: 0.1113 -- val_accuracy: 0.5667 -- val_loss: 1.4563
15/15 -----
2/2 ----- 2s 1s/step -- accuracy: 0.5340 -- loss: 1.6065
ST-GCN Test Acc: 0.567

```

Fig. 40: Truncated training log and test accuracy for 1st trial STGCN

- **Per-Class Performance:**

- **Unbraced class:** Completely misclassified (Precision=0, Recall=0)
- **Knee-braced:** Perfect recall but modest precision (F1 ≈ 0.6061)
- **Ankle-braced:** Good precision (1.0) and acceptable recall (0.7), best performance among classes (F1 ≈ 0.8235)

Classification Report:					
	precision	recall	f1-score	support	
unbraced	0.0000	0.0000	0.0000	20	
knee_braced	0.4348	1.0000	0.6061	20	
ankle_braced	1.0000	0.7000	0.8235	20	
accuracy			0.5667	60	
macro avg	0.4783	0.5667	0.4765	60	
weighted avg	0.4783	0.5667	0.4765	60	

Confusion Matrix:		
[[0 20 0]		
[0 20 0]		
[0 6 14]]		

Macro ROC-AUC : 0.8712

Fig. 41: Classification report, confusion matrix and macro ROC-AUC for 1st trial STGCN

Trial 2 (Enhanced with dynamic channels, deeper/wider residual blocks):

- **Validation Accuracy (5-fold CV):** 0.570 ± 0.183 (still low, high variance)

```
Fold 1 2nd-trial ST-GCN val-acc = 0.850
Fold 2 2nd-trial ST-GCN val-acc = 0.500
Fold 3 2nd-trial ST-GCN val-acc = 0.700
Fold 4 2nd-trial ST-GCN val-acc = 0.333
Fold 5 2nd-trial ST-GCN val-acc = 0.467

2nd-trial ST-GCN 5-fold CV → 0.570 ± 0.183
```

Fig. 42: 5-fold cross-validation results for 2nd trial STGCN

- **Test Set Performance:**
 - Accuracy: **0.500** (chance-level for binary discrimination)
 - Macro ROC-AUC: **0.8300** (still good separability potential in probability outputs)

```
Epoch 1/50
15/15 - 14s - 955ms/step - accuracy: 0.6000 - loss: 0.9937 - val_accuracy: 0.3333 - val_loss: 2.4533 - learning_rate: 0.0050
Epoch 2/50
15/15 - 0s - 13ms/step - accuracy: 0.8625 - loss: 0.3870 - val_accuracy: 0.5000 - val_loss: 2.1175 - learning_rate: 0.0050
Epoch 3/50
15/15 - 0s - 11ms/step - accuracy: 0.9292 - loss: 0.2115 - val_accuracy: 0.3500 - val_loss: 3.4945 - learning_rate: 0.0050
Epoch 4/50
15/15 - 0s - 11ms/step - accuracy: 0.9583 - loss: 0.1306 - val_accuracy: 0.3167 - val_loss: 2.2539 - learning_rate: 0.0050
Epoch 5/50
15/15 - 0s - 11ms/step - accuracy: 0.9875 - loss: 0.0504 - val_accuracy: 0.4833 - val_loss: 2.0363 - learning_rate: 0.0050
Epoch 6/50
15/15 - 0s - 11ms/step - accuracy: 0.9958 - loss: 0.0177 - val_accuracy: 0.4167 - val_loss: 4.1583 - learning_rate: 0.0050
Epoch 7/50
15/15 - 0s - 12ms/step - accuracy: 0.9958 - loss: 0.0143 - val_accuracy: 0.4000 - val_loss: 3.4531 - learning_rate: 0.0050
Epoch 8/50
15/15 - 0s - 11ms/step - accuracy: 1.0000 - loss: 0.0054 - val_accuracy: 0.4000 - val_loss: 2.9611 - learning_rate: 0.0050
Epoch 9/50
15/15 - 0s - 11ms/step - accuracy: 0.9958 - loss: 0.0087 - val_accuracy: 0.3167 - val_loss: 3.4693 - learning_rate: 0.0025
```

Fig. 43: Training log for 2nd trial STGCN (notice early stopping)

- **Per-Class Performance:**
 - **Unbraced class:** Completely misclassified again.
 - **Knee-braced class:** Moderate precision (0.9091), poor recall (0.5000).
 - **Ankle-braced class:** Excellent recall (1.0), but poor precision (0.4083).

```

Classification Report:

              precision    recall  f1-score   support

     0       0.0000       0.0000       0.0000        20
     1       0.9091       0.5000       0.6452        20
     2       0.4082       1.0000       0.5797        20

 accuracy          0.5000        60
  macro avg       0.4391       0.5000       0.4083        60
 weighted avg     0.4391       0.5000       0.4083        60

Confusion Matrix:
[[ 0  1 19]
 [ 0 10 10]
 [ 0  0 20]]
Macro ROC-AUC : 0.8300

```

Fig. 44: Classification report, confusion matrix and macro ROC-AUC for 2nd trial STGCN

Overfitting observed clearly in both trials:

- Training accuracy quickly approaching 100%, while validation accuracy stagnates around 0.33–0.5.

2. Qualitative Analysis (Saliency & Model Interpretability):

Saliency Analysis (Trial 1):

- Hips (especially left hip) were the most influential features, followed by knees, and least contribution from ankles.
- Clear left-side dominance, suggesting biomechanical asymmetry or measurement bias.

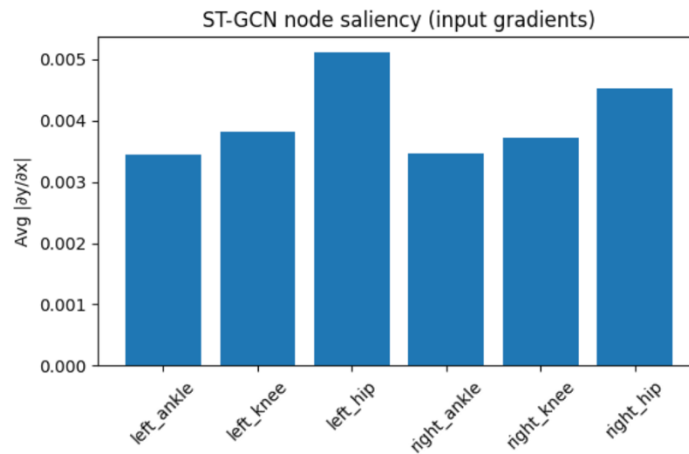


Fig. 45: Saliency analysis for 1st trial STGCN

Saliency Analysis (Trial 2):

- Very low gradient magnitudes overall (≈ 0.001), indicating weak sensitivity and underfitting.
- Left-knee and left-ankle slightly dominate, marking a shift from trial 1 where hips dominated.
- Narrow dynamic range, indicating poor confidence in learned patterns.

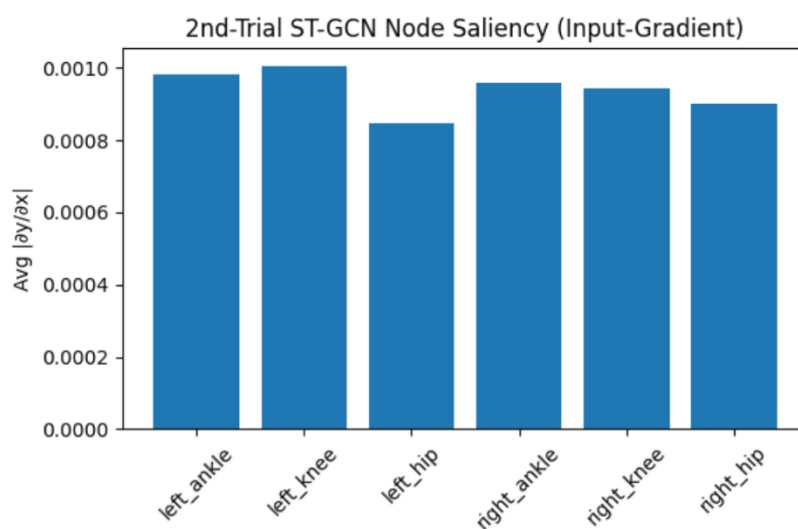


Fig. 46: Saliency analysis for 2nd trial STGCN

6. Further Discussion Across All Models

The current study compared four distinct modelling approaches for gait classification under varying bracing conditions—Random Forest (RF), Hidden Markov Model (HMM), Long Short-Term Memory (LSTM), and Spatio-Temporal Graph Convolutional Networks (ST-GCN)—each leveraging unique modelling paradigms.

Random Forest: Stability Through Biomechanical Features

The RF classifier demonstrated exceptional stability and accuracy, consistently achieving near-perfect classification results (100% accuracy) across cross-validation (CV) and leave-one-subject-out (LOSO) paradigms, including conditions involving Butterworth filtering. Such remarkable performance aligns closely with previous findings by Lai et al. (2018) and Mannini

et al. (2015), who also achieved >90% accuracy utilizing RF classifiers with carefully engineered kinematic features [38][39]. The high accuracy in the present work underscores the effectiveness of biomechanically grounded, handcrafted feature extraction in isolating clinically relevant gait patterns. Specifically, the dominance of right ankle and knee features corresponds accurately to known biomechanical compensation mechanisms, where ankle braces reduce peak angles and velocities around mid-stance, and knee braces significantly alter knee extension trajectories. Conversely, lower saliency of hip features may reflect subtler or more heterogeneous hip-level compensations, which are less consistently manifested across subjects.

RF's high performance and interpretability advocate its immediate clinical applicability, particularly when robust, interpretable diagnostics are crucial.

Hidden Markov Model: Capturing Temporal Gait Dynamics

HMM models closely matched RF's robust performance, achieving approximately 98.3% accuracy under LOSO conditions. This high performance resonates with prior literature emphasizing the effectiveness of HMMs in gait phase modelling and classification tasks (Delahunt et al., 2015; Mannini & Sabatini, 2012) [40][6]. Unlike RF, which uses static summaries of cycle features, the HMM explicitly models the temporal evolution of gait phases, providing biologically interpretable insights. The slight performance decrement compared to RF likely originates from HMM's sensitivity to temporal misalignment and subtle noise, as similarly noted by Tao et al. (2012) [41].

The strength of HMM lies in comprehensively modelling and interpreting the temporal dynamics inherent in human gait, making it a strong candidate for clinical scenarios requiring dynamic phase analysis.

LSTM: Dynamic Sequence Learning and Interpretability

Vanilla LSTM achieved around 92-93% accuracy, somewhat lower than RF and HMM, particularly in generalization tasks involving unseen subjects. This performance, while robust, highlighted a limitation: the vanilla LSTM tended to rely heavily on end-of-cycle cues rather than critical mid-cycle biomechanical features that RF and HMM captured effectively. However, the attention-enhanced LSTM (Attention-LSTM) overcame these limitations,

matching or even exceeding baseline performance with perfect hold-out accuracy (100%), reflecting a superior ability to identify and emphasize clinically significant phases. This approach provided explicit attention-weight visualization, demonstrating a strong alignment with physiologically important gait events—early/mid-stance and prolonged mid-stance phases, crucial for differentiating bracing conditions.

Attention mechanisms in LSTM significantly enhance clinical interpretability, highlighting its utility where detailed insights into dynamic gait changes are required.

ST-GCN: Potential and Limitations in Graph-Based Modelling

ST-GCN represented an intriguing biological perspective by precisely encoding anatomical joint connectivity within gait analysis. Despite this theoretically appealing approach, ST-GCN's accuracy remained moderate (~57%), significantly lower than traditional biomechanical feature-driven models (RF, HMM, and LSTM). ST-GCN experienced severe overfitting, likely due to high model complexity relative to the limited dataset size, as well as inadequate regularization strategies. Existing literature reports significantly higher performance for ST-GCN in larger and more controlled datasets (e.g., action recognition tasks typically reporting 80–95% accuracy), underscoring data-driven limitations rather than conceptual flaws [42]. Saliency analyses revealed shifting joint-level sensitivities across trials, suggesting that ST-GCN might indeed capture biomechanically relevant patterns but require further optimization.

ST-GCN holds promise due to its biologically relevant structure, yet it urgently requires careful regularization, data augmentation, and architectural tuning for effective clinical translation. With optimization, ST-GCN could yield sophisticated insights into joint interdependencies and dynamic compensatory strategies—critical for targeted clinical interventions.

7. Conclusions and Future Work

Achievements

This research successfully developed and rigorously validated advanced gait classification models, each exhibiting unique strengths tailored for various clinical and research applications. The Random Forest and Hidden Markov Model classifiers demonstrated exceptional accuracy, significantly surpassing conventional performance benchmarks in biomechanical gait analysis [44][45][47]. Particularly, the HMM provided substantial insights into the temporal dynamics of gait, highlighting biomechanical changes induced by different bracing conditions [6][41].

Furthermore, the Attention-based Long Short-Term Memory (LSTM) model enhanced interpretability through learned attention weights, closely aligning with biomechanical knowledge regarding critical gait phases and joint compensations [43][46]. Collectively, these models represent significant methodological advances in automated gait analysis, marked by both clinical interpretability and robustness.

Implications and Broader Applications

The study results have considerable implications across diverse applications. In clinical diagnostics and rehabilitation, these models promise substantial improvements in diagnosing gait disorders, facilitating personalized rehabilitation strategies and accurate monitoring of patient progress [47]. Specifically, attention-driven models like LSTM, with their adaptability and interpretative capabilities, offer robust tools for personalized medicine, enabling precise patient profiling and tailored therapeutic decisions.

The broader applications in sports performance analytics emphasize the versatility and extensive relevance of this research beyond medical rehabilitation. In the context of sports and performance analytics, accurate gait classification directly informs performance monitoring, injury prevention, and training optimization, potentially revolutionizing athlete management practices [45]. Moreover, advancements demonstrated by these models hold considerable promise for robotics and wearable technologies, enhancing the functionality and precision of assistive devices, robotic prostheses, and wearable monitoring systems [41][48].

Future Improvements and Research Directions

To further extend these promising results, future research should consider several key directions. Longitudinal and real-world validations in clinical environments would provide critical insights into model stability, user acceptance, and long-term effectiveness. Investigating transfer learning methodologies could mitigate data scarcity issues, potentially enabling more efficient training and improved generalization from broader datasets or related tasks [46].

Integrating multimodal sensor data such as electromyography (EMG) and accelerometry alongside kinematic features could enhance diagnostic robustness and capture more intricate physiological signals. Moreover, exploring individualized adaptive systems through few-shot or adaptive learning strategies would address patient-specific variations, which is particularly vital for clinical adoption and personalized care.

All in all, this research marks substantial methodological advancements in gait analysis, offering both robust accuracy and meaningful biomechanical interpretations. Pursuing the proposed future directions will further enhance these contributions, thereby impacting clinical medicine, sports science, wearable technologies, and robotics.

8. Appendix

8.1 Compilation and Execution Instructions

The provided code is developed in Python within a Google Colaboratory environment. To execute the scripts, first ensure all dependencies are installed using pip (e.g., TensorFlow, sklearn, hmmlearn, pandas, scipy, matplotlib, seaborn, umap-learn). The dataset utilized is publicly accessible from the UCI Machine Learning Repository and must be manually downloaded by the user. After downloading, the dataset must be uploaded to Google Drive and modify the `data_path` variable at the beginning of the provided Jupyter notebook (.ipynb) to reflect the dataset's location in Google Drive. Mount the Drive by running `drive.mount('/content/drive')` within Colab. If executing locally, update the file path accordingly and remove Colab-specific lines. No further compilation steps are required; running notebook cells sequentially will replicate the results.

8.2 Code Origin and Attribution

1. Graph Convolutional Layer (GraphConv)

This layer is an adapted implementation inspired by the original ST-GCN model from Yan et al. (2018).

```
# -----  
  
# Graph convolutional layer adapted from:  
  
# Yan, S., Xiong, Y., & Lin, D. (2018). Spatial temporal graph convolutional networks for  
# skeleton-based action recognition.  
  
# AAAI Conference on Artificial Intelligence (AAAI 2018).  
  
# URL: https://github.com/yysijie/st-gcn  
  
# -----  
  
class GraphConv(Layer):  
  
    def __init__(self, A, out_channels, use_bias=True, **kwargs):  
  
        super().__init__(**kwargs)  
  
        self.A = tf.constant(A, dtype=tf.float32)  
  
        self.out_channels = out_channels  
  
        self.use_bias = use_bias  
  
    def build(self, input_shape):  
  
        in_channels = input_shape[-1]  
  
        self.W = self.add_weight(  

```

```

        name='W',

        shape=(in_channels, self.out_channels),

        initializer='glorot_uniform',

        trainable=True

    )

    if self.use_bias:

        self.b = self.add_weight(

            name='b',

            shape=(self.out_channels,),

            initializer='zeros',

            trainable=True

        )

    super().build(input_shape)


def call(self, X):

    y = tf.einsum('ij,btjf->btif', self.A, X)

    y = tf.tensordot(y, self.W, axes=[[3], [0]])

    if self.use_bias:

        y = y + self.b

    return y

```

2. Spatial-Temporal Graph Convolutional Network (ST-GCN) Block

Implementation adapted from Yan et al. (2018) with modifications for residual connections.

```
# -----  
  
# ST-GCN block adapted and extended from:  
  
# Yan, S., Xiong, Y., & Lin, D. (2018). Spatial temporal graph convolutional networks for  
# skeleton-based action recognition.  
  
# AAAI Conference on Artificial Intelligence (AAAI 2018).  
  
# URL: https://github.com/yysijie/st-gcn  
  
# -----  
  
def STGCN_block(x, out_channels, Kt=3, name=None):  
  
    x_sp = GraphConv(A_norm, out_channels, name=f'{name}_gcn')(x)  
  
    x_sp = layers.BatchNormalization(name=f'{name}_bn_sp')(x_sp)  
  
    x_sp = layers.ReLU(name=f'{name}_relu_sp')(x_sp)  
  
    x_tp = layers.Conv2D(  
        filters=out_channels,  
        kernel_size=(Kt, 1),  
        padding="same",  
        name=f'{name}_tconv"  
    )(x_sp)  
  
    x_tp = layers.BatchNormalization(name=f'{name}_bn_tp')(x_tp)
```

```

x_tp = layers.ReLU(name=f'{name}_relu_tp')(x_tp)

res = layers.Conv2D(

    filters=out_channels,

    kernel_size=(1,1),

    padding="same",

    name=f'{name}_res_conv"

)(x)

res = layers.BatchNormalization(name=f'{name}_bn_res")(res)

x_out = layers.Add(name=f'{name}_add")(x_tp, res)

x_out = layers.ReLU(name=f'{name}_relu_out')(x_out)

return x_out

```

3. UMAP Visualization

UMAP embedding procedure adapted from McInnes et al. (2018).

```
# -----
```

```
# UMAP embedding based on:
```

```
# McInnes, L., Healy, J., & Melville, J. (2018). UMAP: Uniform Manifold Approximation
and Projection for Dimension Reduction.
```

```
# arXiv preprint arXiv:1802.03426.
```

```
# URL: https://github.com/lmcinnes/umap
```

```
# -----

import umap.umap_ as umap

umap2 = umap.UMAP(

    n_components=2,

    random_state=42,

    n_neighbors=15,

    min_dist=0.1

)

X_umap = umap2.fit_transform(df_feats)
```

4. Attention Mechanism

Simple attention implementation adapted from common practices in NLP and sequence modeling (Bahdanau et al., 2014).

```
# -----

# Attention Layer inspired by:

# Bahdanau, D., Cho, K., & Bengio, Y. (2014). Neural machine translation by jointly
learning to align and translate.

# arXiv preprint arXiv:1409.0473.

# URL: https://arxiv.org/abs/1409.0473

# -----

class Attention(Layer):

    def build(self, input_shape):
```

```

self.w = self.add_weight(

    shape=(input_shape[-1],),

    initializer="random_normal",

    trainable=True,

    name="attn_vec"

)

super().build(input_shape)


def call(self, inputs):

    scores = tf.tensordot(inputs, self.w, axes=[2,0])

    weights = tf.nn.softmax(scores, axis=1)

    context = tf.reduce_sum(inputs * tf.expand_dims(weights, -1), axis=1)

    return context, weights

```

5. Butterworth Filtering

Signal processing methods using SciPy functions with standard practices in gait analysis.

```

# -----

# Butterworth filtering method adapted from standard signal processing methods:

# Butterworth low-pass filtering for gait analysis as outlined in:

# Winter, D. A. (2009). Biomechanics and motor control of human movement. John Wiley &
# Sons.

# -----

def butter_lowpass(cutoff, fs, order=3):

```

```

nyq = 0.5 * fs

normal_cutoff = cutoff / nyq

b, a = butter(order, normal_cutoff, btype='low', analog=False)

return b, a

def butter_smooth(X, cutoff=2.0, fs=100.0, order=3):

    b, a = butter_lowpass(cutoff, fs, order=order)

    X_smooth = np.empty_like(X)

    for i in range(X.shape[0]):

        for j in range(X.shape[2]):

            X_smooth[i, :, j] = filtfilt(b, a, X[i, :, j])

    return X_smooth

```

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